भारतीय सांकेतिक विज्ञान संस्थ हैदराबाद
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

# Part B.2 - MOSCAP

*Q: Most Important Interface – in Humankind?*

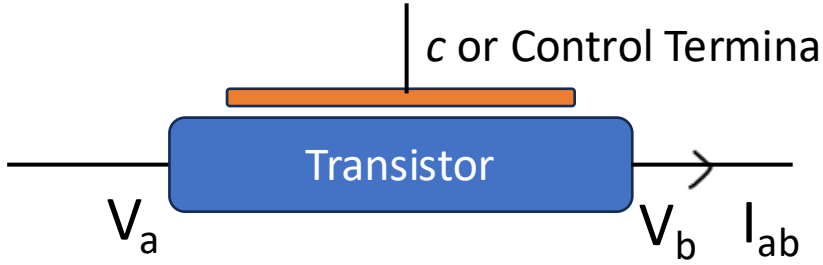
Topics: *Importance of oxide/semiconductor Interface, MOSCAP Ideal Operating Conditions, $Q_s$ vs $\psi_s$, C-V Characteristics, Non-Idealities in MOSCAP, Extraction of parameters*

Semiconductor Device Fundamentals

EE, IIT Hyderabad

Shubhadeep B (shubhadeep@ee at iith)

# What is a Transistor? BJT from *pn* Junc




Transfer-Resistance: 3T (a,b,c)

$I_{ab}$ independent of $V_{ab}$ (A Current Source)




What controls $I_{ab}$? $V_c$ or $I_c$

Current Controlled Current Source

Ex. BJT,HBT

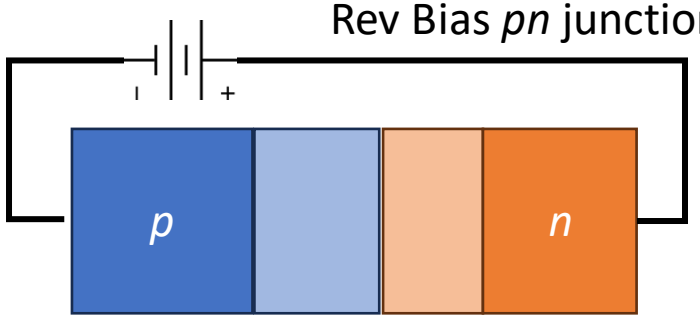
Voltage Controlled Current Source

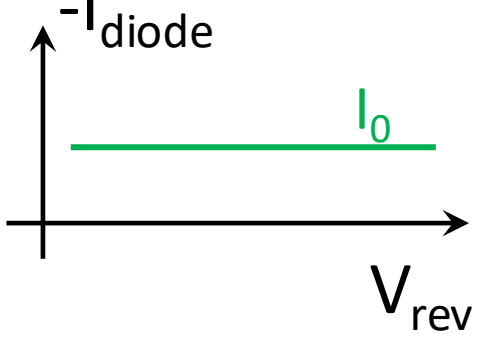
Ex. JFET, MOSFET

## Construction of BJT

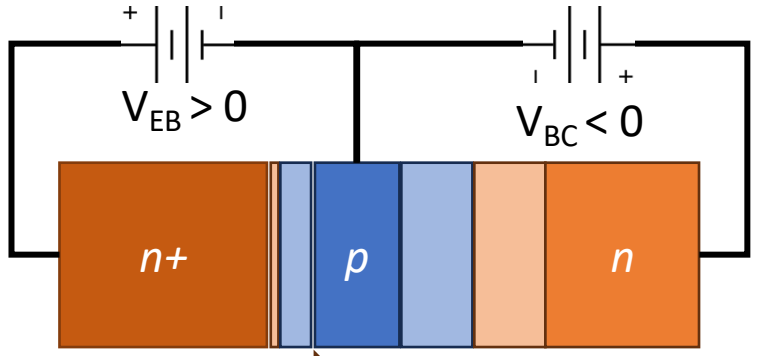
We already know a constant current source

Rev Bias *pn* junction




$I_{0,rev} \propto n_p$ , $p_n$ To regulate $I_{0,rev}$ min. carr. injection (?)




This is a n/p/n BJT!
n+ Emitter
p- Base (Thin)
n Collector

*Inject e- into p –base*

*$\Delta n_p \propto exp(V_{EB}/V_t)$*

*Nearly all e- swept into collector by $E_{CB}$ minimal Recomb.*

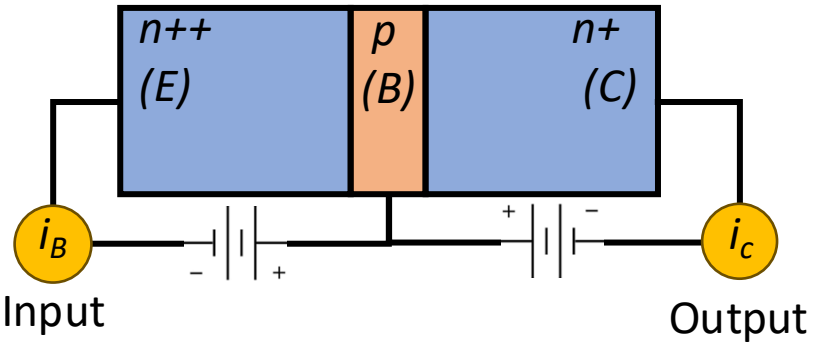
# Transition from BJTs to MOSFETs

***Q: Why did we move from BJTs to MOSFETs ?***

BJTs developed much earlier – but now restricted to niche applications

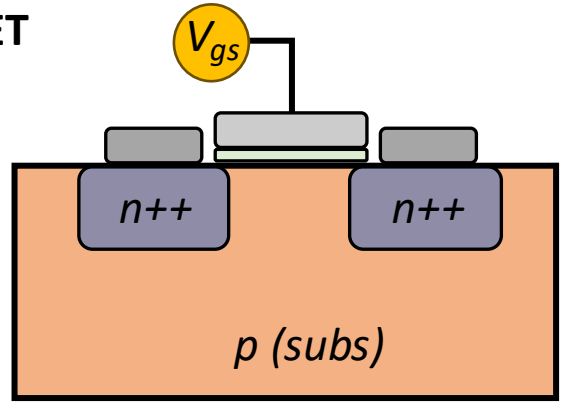
MOSFETs capture nearly 99.99% of all transistor device

**BJT**




| 1. $R_{out}/R_{in}$ | 2. Static Loss | 3. *Voltage* and *Area* |
|---|---|---|
| $= i_c/i_b = \beta \sim 100$<br>Limited Fanout ☹ | ☹ $J_{rev,\ sat} \sim \mu A$ | Poor! ☹<br>Vertical Structure, $V_{BE}$ |

**MOSFET**

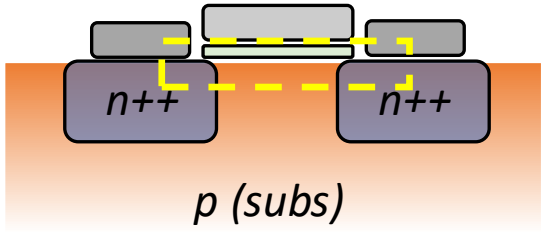



| | | |
|---|---|---|
| $= I_d/I_g \sim 10^6$<br>Large Fanout ☺ | ☺ $I_g$, $I_{d_off} \sim$ pA-nA | Great! ☺<br>Lateral Structure, $V_{th}$ |

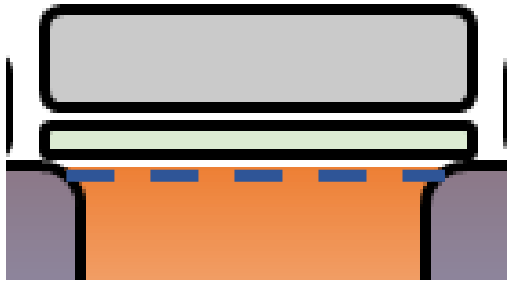
**While BJT high frequency low noise Analog applications:**
**MOSFETs rule in logic, digital and most other applications**

# MOSCAP – Structure and Importance

***Q: MOSFETs - conceptualized in 1920s-30's – Why not developed till 1960's ?***




*MOSCAP: Functional Junc of MOSFET*

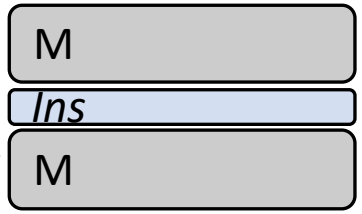


*"The devil is in the interface !"*
*$D_{it}$ ~$10^{13}$ $cm^{-2}$ -> $D_{it}$ ~$10^{10}$ $cm^{-2}$*
*@ Oxide/Semiconductor Interface (growth and anneal conditions)*

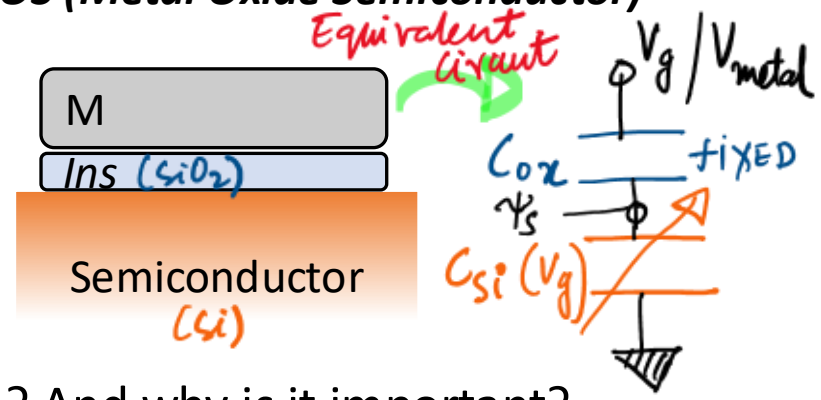
*Ge carr. Mobility*
*Ge -> GeO (water soluble)*

***MIM (Metal Insulator Metal)*** ***MOS (Metal Oxide Semiconductor)***

Symmetric,
+Q comp by –Q
C vs V flat




*Vs.*



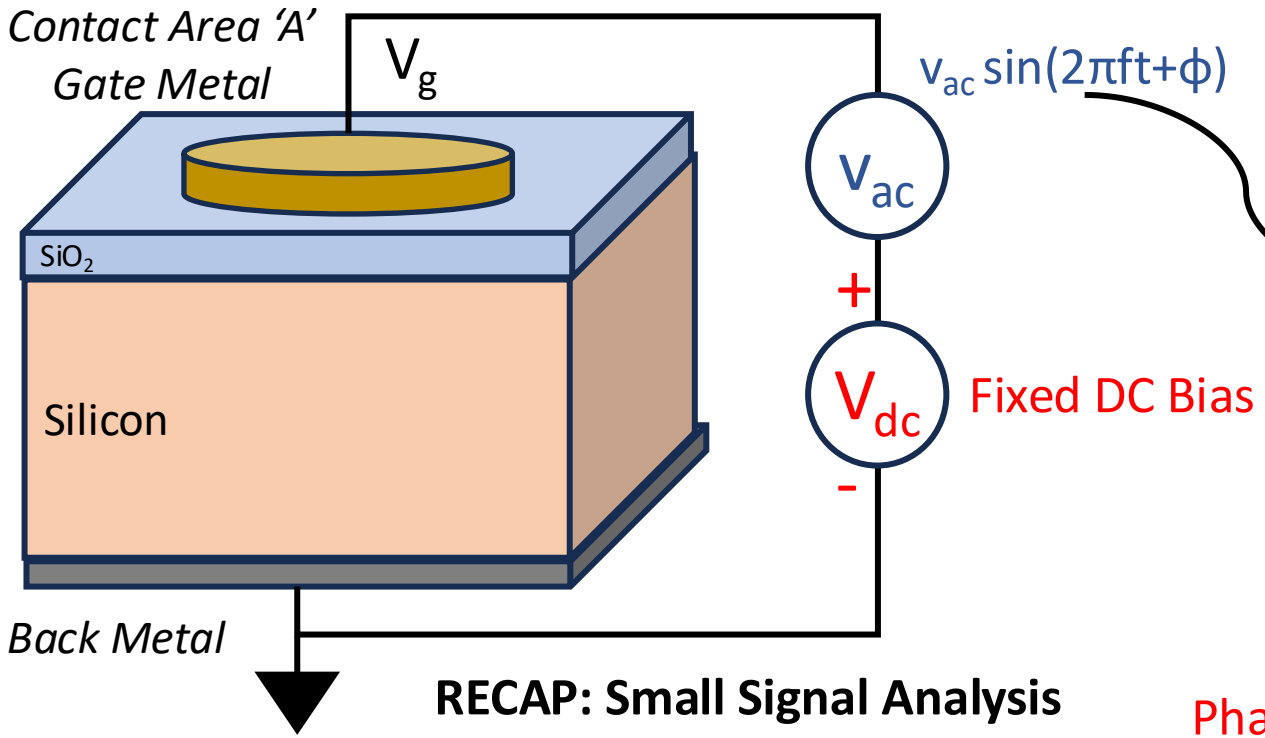

Why is $C_{si}$ function of $V_g$? And why is it important?
A: Mobile (n,p) move & Fixed ($N_A$-, $N_D$+) which uncover with $V_g$
=> Surface & Depletion Charge in Si in response to $V_g$

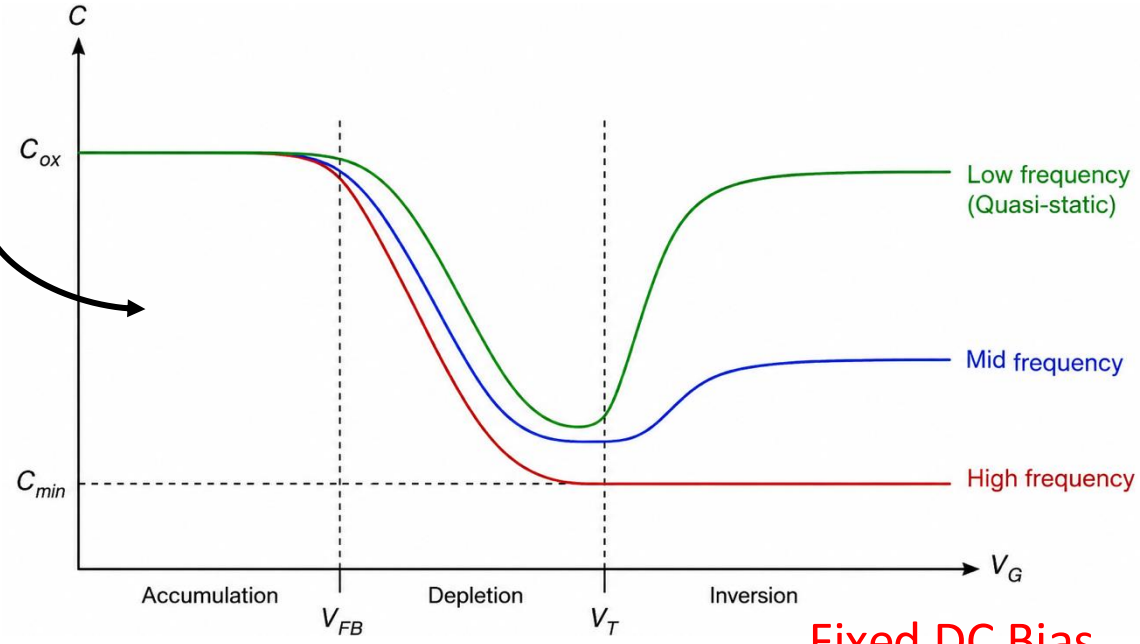
MOSCAP C-V: Most Versatile Diagnostic/ Parameter Extraction tool.

INTRISIC
- Doping: $N_A$,$N_D$
- $C_{ox}$, $t_{ox}$, $\varepsilon_{ox}$
- Quality/Purity /Defects in Si

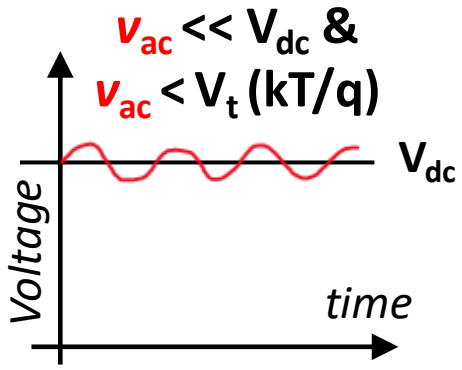
EXTRINSIC
- Interface Traps
- Fixed Charges

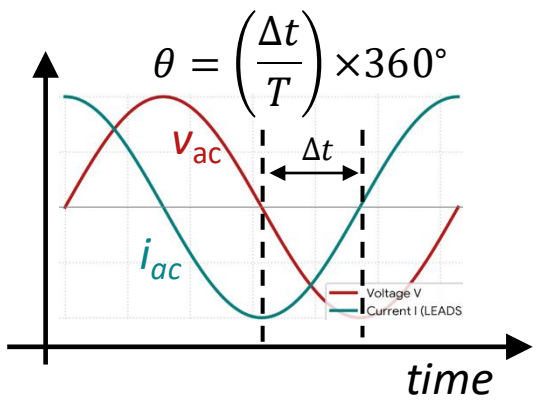
# MOSCAP – CV How is it measured?



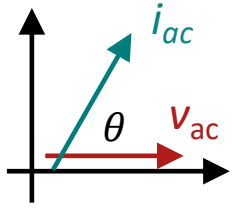




## RECAP: Small Signal Analysis




Phasor




$$\tan\theta = \frac{X_C}{R_L} = \frac{1}{j\omega C R_L}$$

MOSCAP C-V: Most Versatile Diagnostic/ Parameter Extraction tool.

INTRISIC

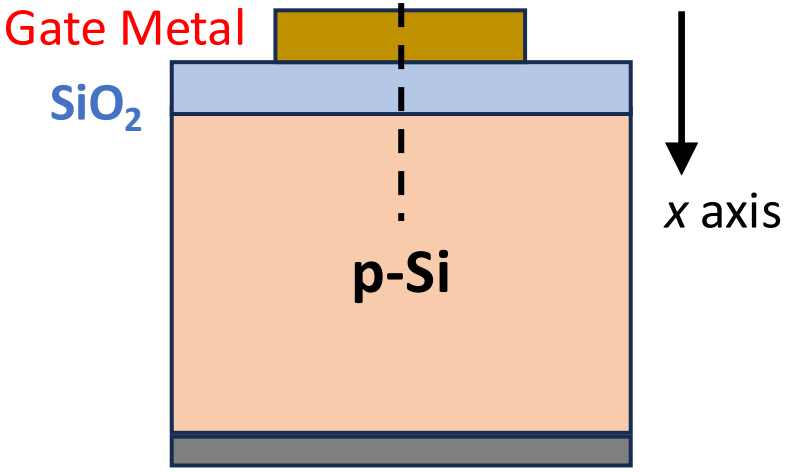
- Doping: $N_A$,$N_D$
- $C_{ox}$, $t_{ox}$, $\varepsilon_{ox}$
- Quality/Purity /Defects in Si

EXTRINSIC

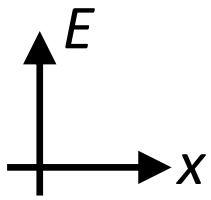
- Interface Traps
- Fixed Charges

# Let's start: *n*-MOSFET, p-Substrate

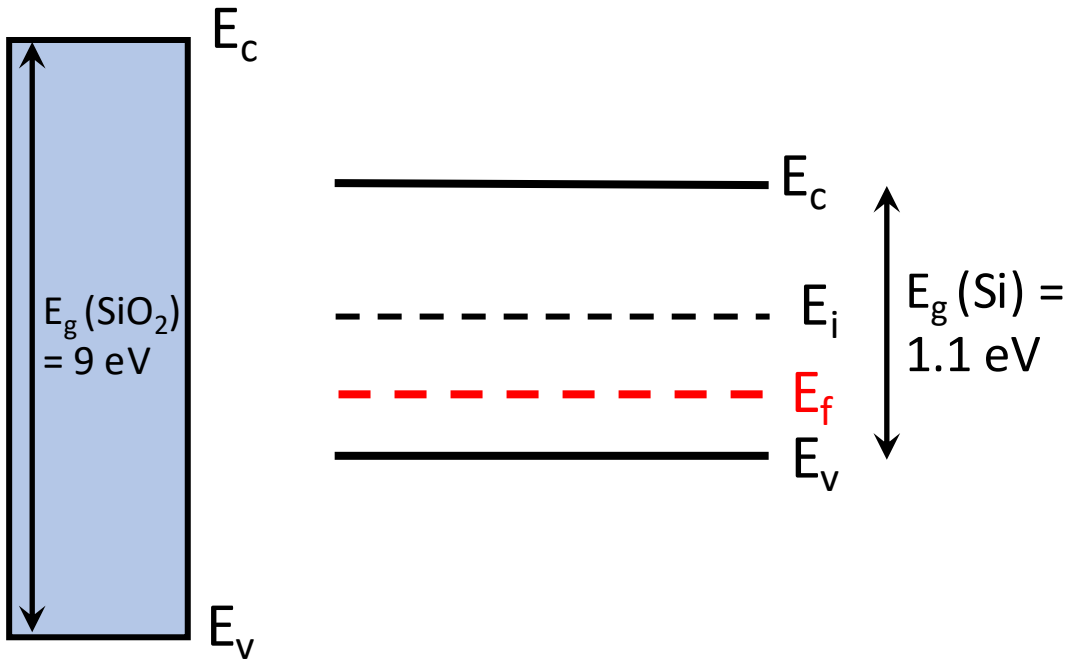
Cross-section – All the action is happening




Back Metal – Ohmic, transparent contact



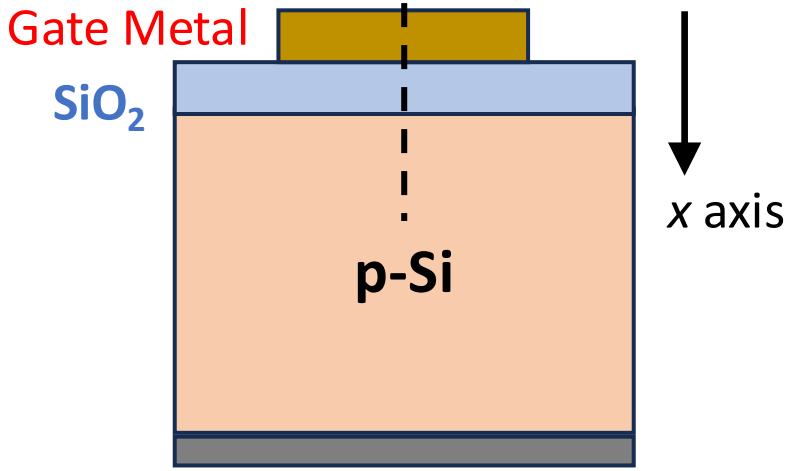

Band Diagram (not to scale)




$SiO_2$: Perfect Insulator

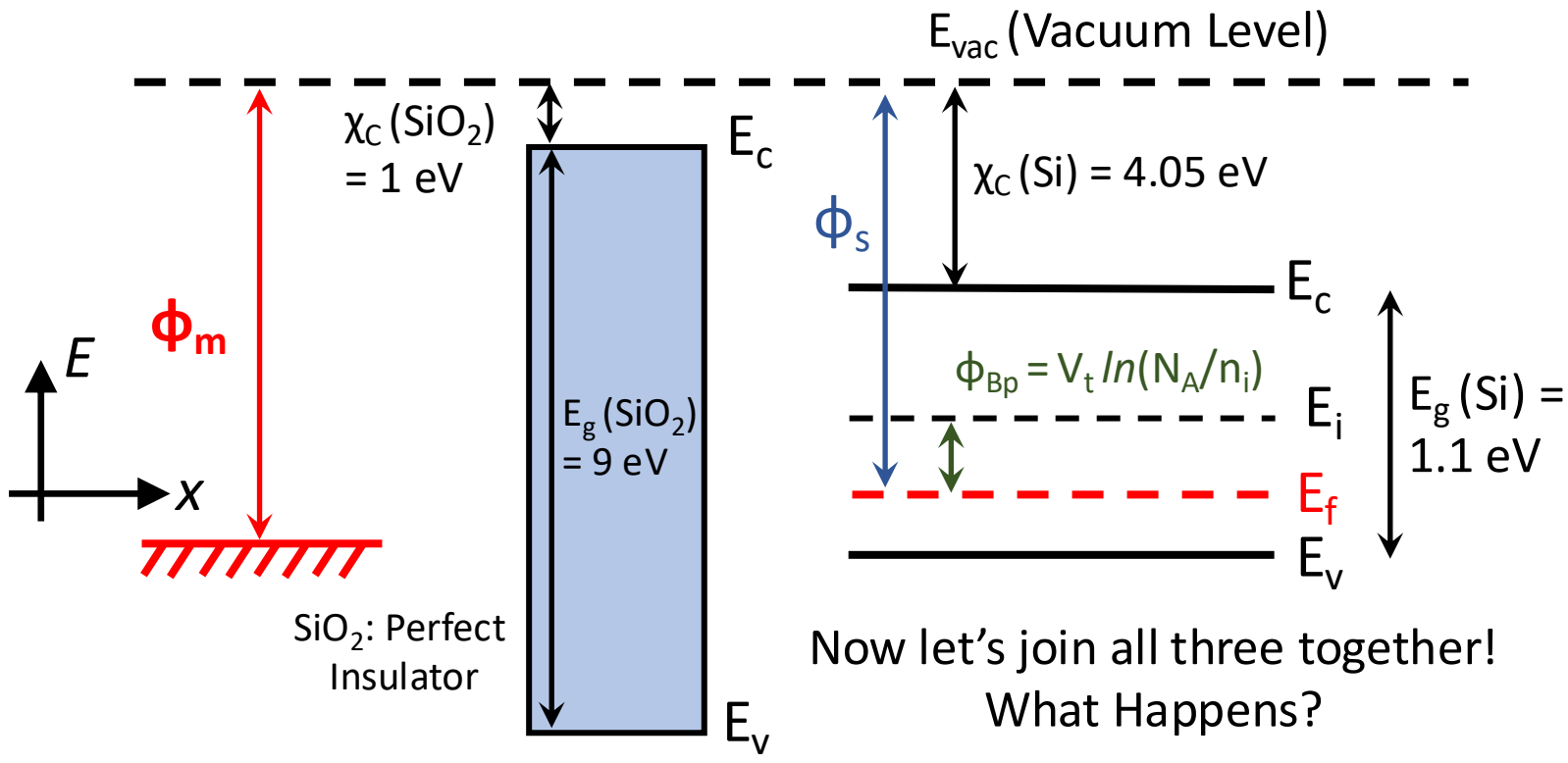
# Let's start: *n*-MOSFET, p-Substrate

Cross-section – All the action is happening




Back Metal – Ohmic, transparent contact

Band Diagram (not to scale)




Now let's join all three together! What Happens?

New Concepts useful for MOSCAP:

1) $E_{vac}$/Vacuum Level
Free Electrons (no lattice Potential) Reference level

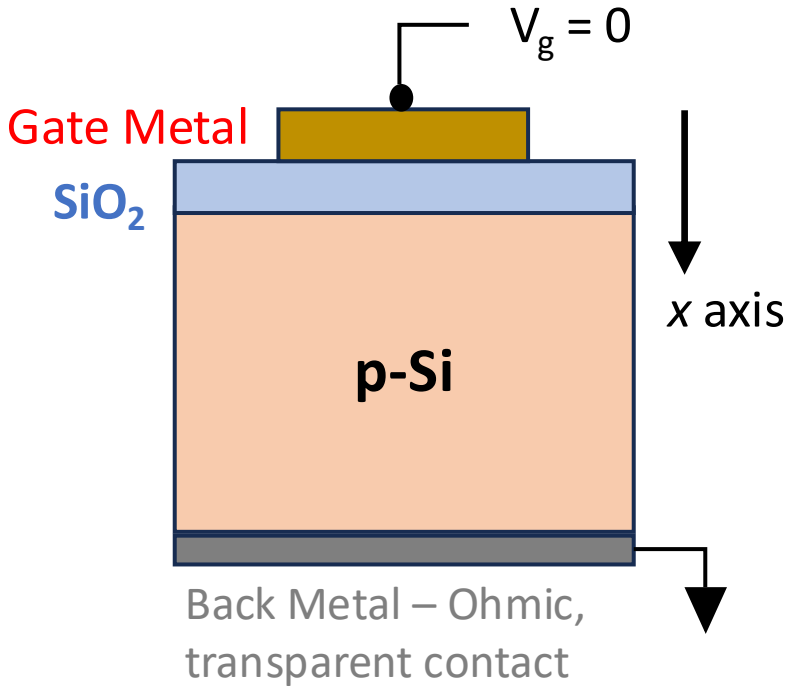
2) Electron Affinity ($\chi_C$)
energy required for conduction e- to "escape" to $E_{vac}$

3) Metal Work Function ($\phi_m$)
energy required for valence e- in metal to "escape" to $E_{vac}$ ; $E_{vac}$- $E_{f,metal}$

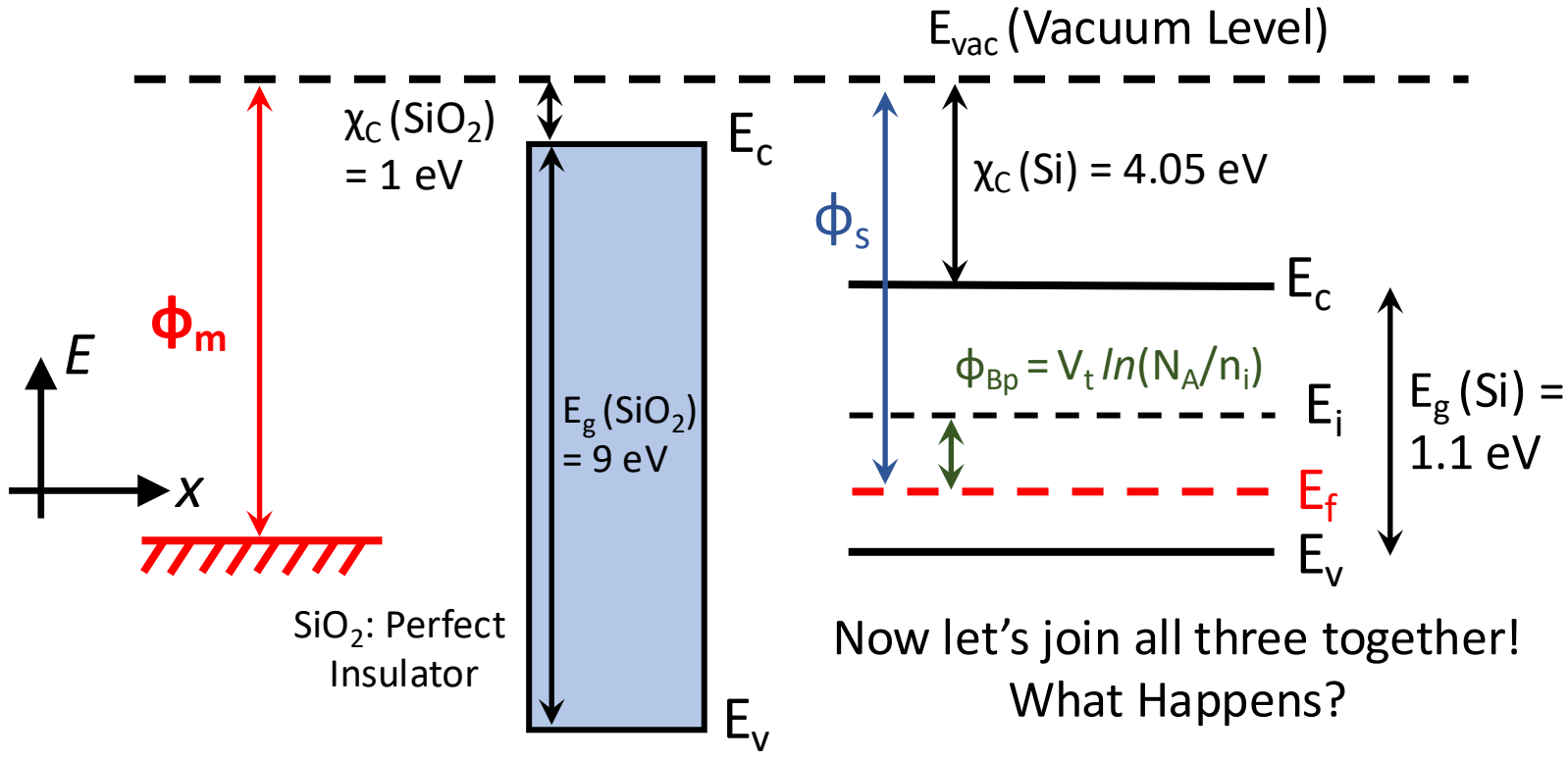
4) Semiconductor Work Function ($\phi_s$)
$E_{vac}$- $E_{f,semicon}$
$= \chi_C + E_g - (E_f - E_v)$
$= \chi_C + E_g - V_t \ln(N_v/N_A)$

5) Body Potential ($\phi_B$) Semicon
$E_i - E_f = V_t \ln(N_A/n_i)$

# $V_g = 0$; Ideal MOSCAP assumptions




Band Diagram (not to scale)



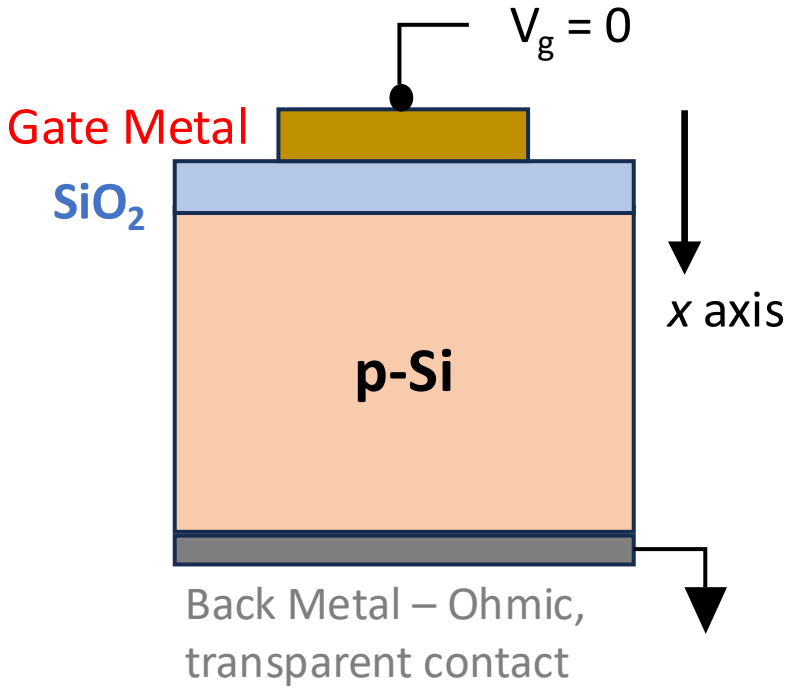

Now let's join all three together! What Happens?

Very Important:
In MOSCAP No charge transfer b/w Metal and Semicon => perfect Insulator in b/w {unlike diode}

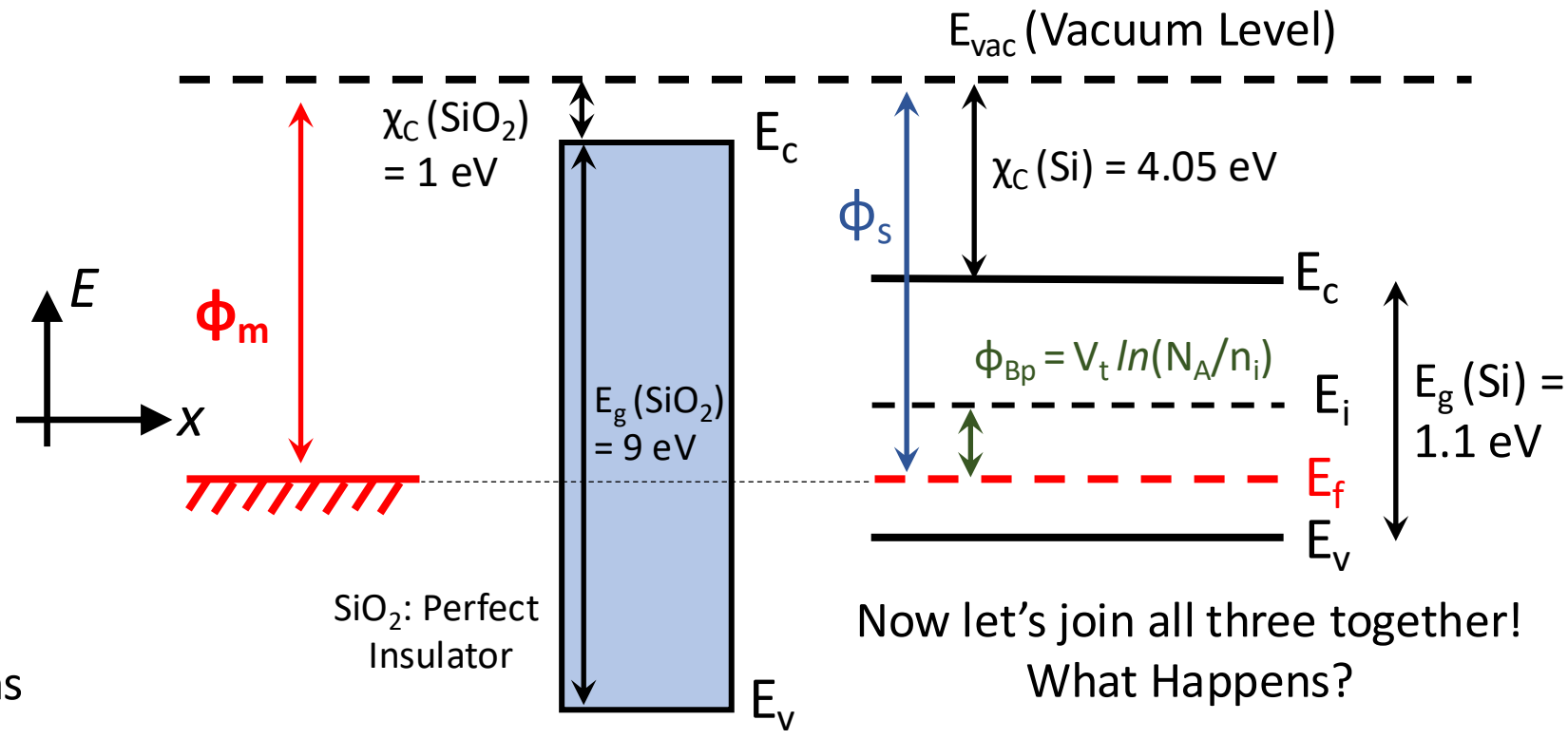
Floating condition unpredictable in MOSCAP must be avoided (during measurements)

Gate and Bottom Metal Connected (Equipotential)
Here, $V_{back}$ = GND; $V_g = 0$
{Equilibrium Band Diagram}

# $V_g = 0$; Ideal MOSCAP assumptions




Band Diagram (not to scale)




Two Simplifying Assumptions for ease of analysis:

1) $\phi_m - \phi_s = \phi_{ms} = 0$
i.e. No charge transfer b/w Metal and Semicon

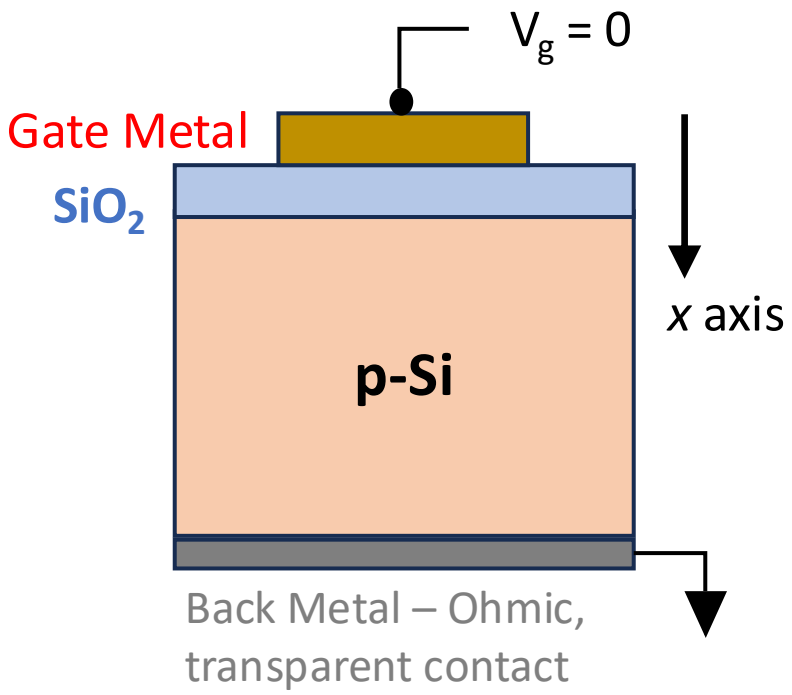
2) Ideal Insulator:
No fixed/trapped charges &
No Leakage through oxide

Now let's join all three together!
What Happens?

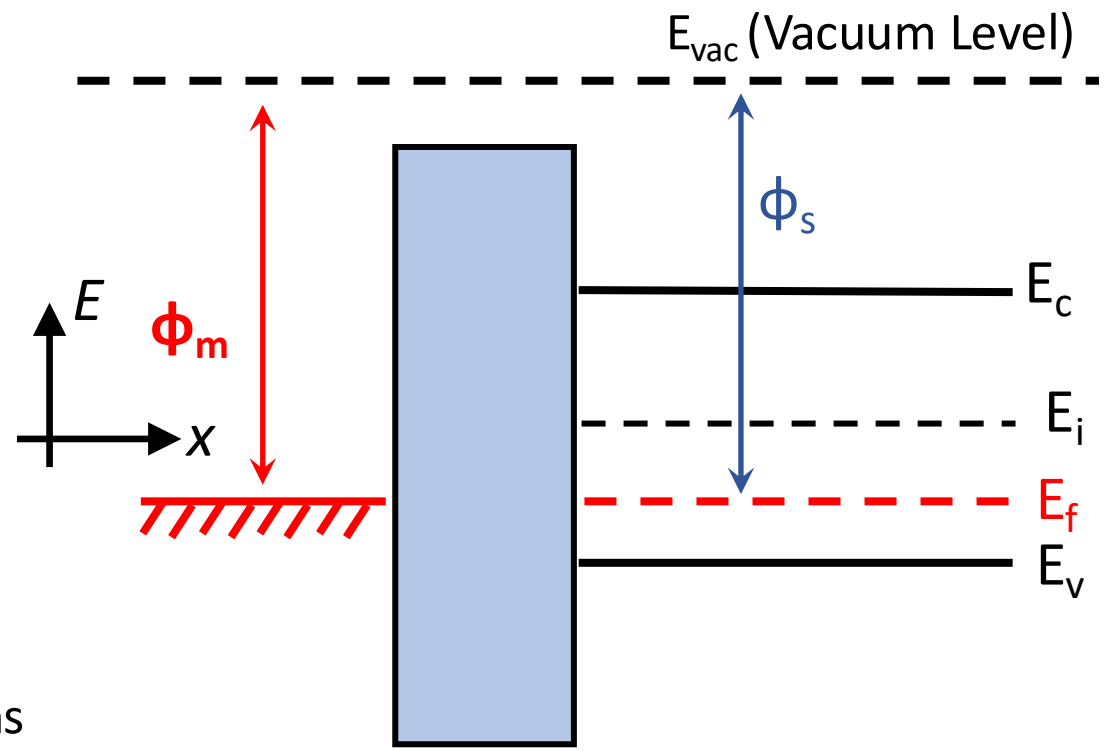
Implication of these assumptions:
Flat Band Voltage $V_{fb} = 0$
Or, Bands are flat, $E_{field} = 0$ at $V_g = 0$

Departures from ideal assumptions: Later $V_{fb} \neq 0$

# $V_g = 0$; Ideal MOSCAP assumptions




~~Equilibrium~~ Band Diagram (Idealized)




$V_{fb} = 0$ because $\phi_{ms} = 0$
A clean slate to start proceedings:
i.e. Apply $V_g$

Two Simplifying Assumptions for ease of analysis:

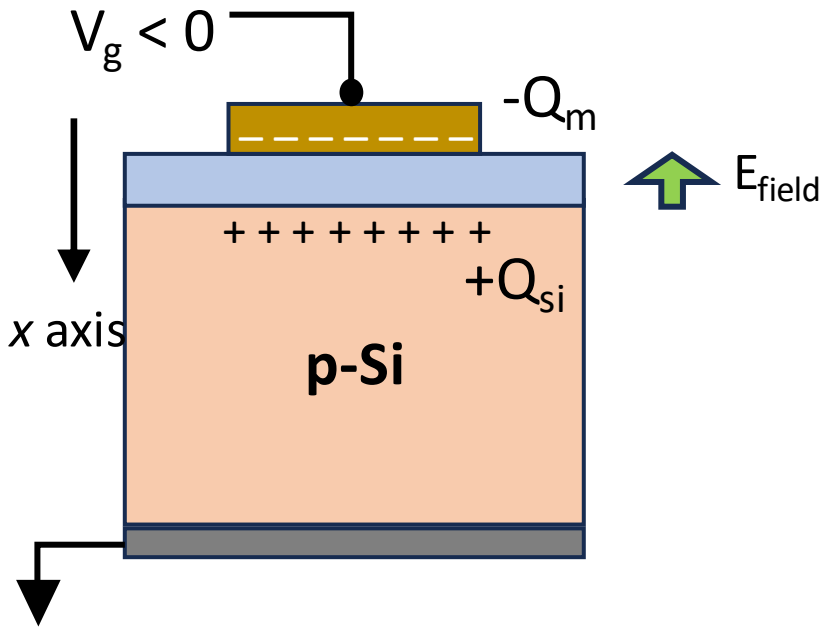
1) $\phi_m - \phi_s = \phi_{ms} = 0$
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Implication of these assumptions:
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Departures from ideal assumptions: Later $V_{fb} \neq 0$

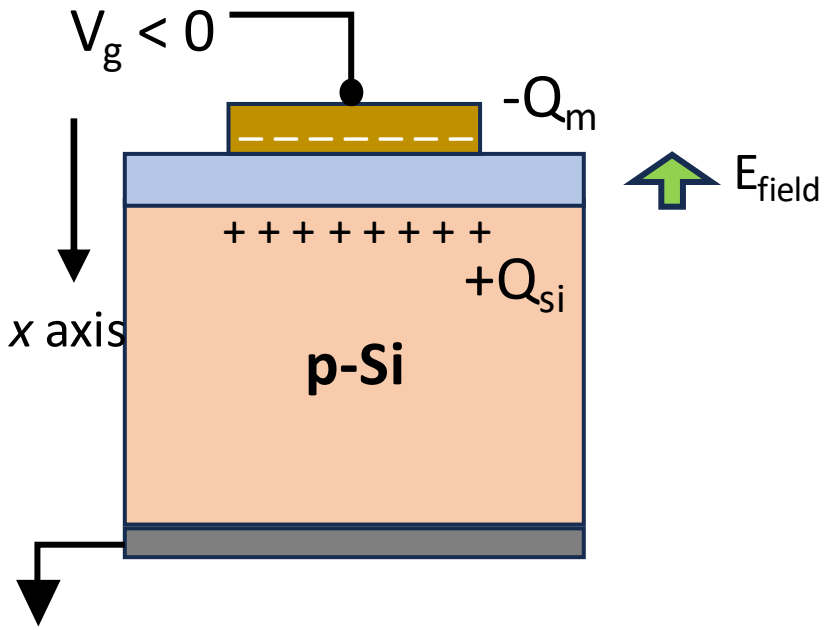
# $V_g < 0$ or Accumulation




$V_g < 0$; $-Q_m$ on metal
Compensated by $+Q_{si} = -Q_m$
In *p*-Si: p+, n-, $N_A$- : Only p+

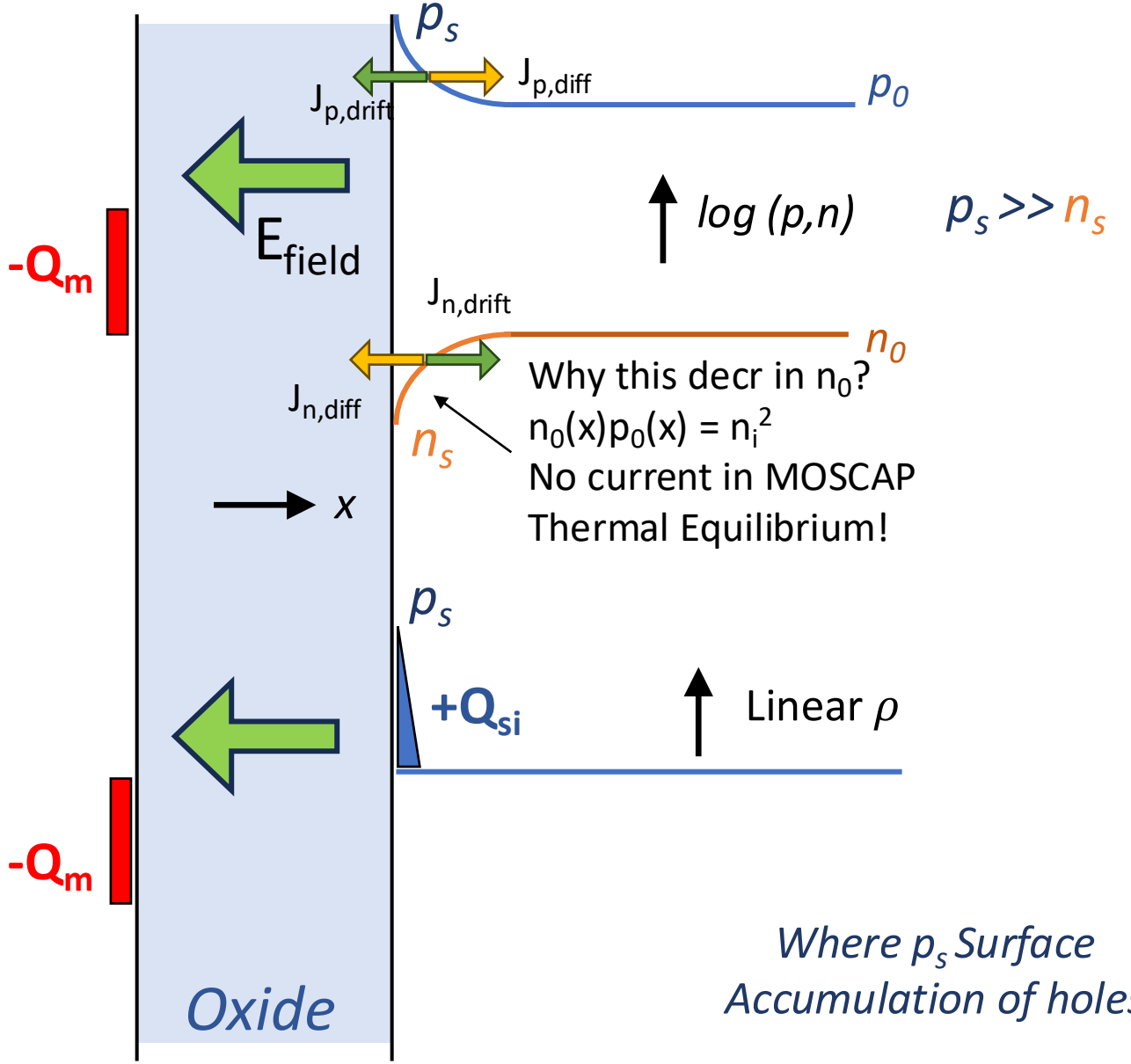
Let's take $N_A = 10^{15}$ cm$^{-3}$,
$p_0 = 10^{15}$ cm$^{-3}$; $n_0 = 10^5$ cm$^{-3}$

# $V_g < 0$ or Accumulation $p_s > p_0 >> n_s$



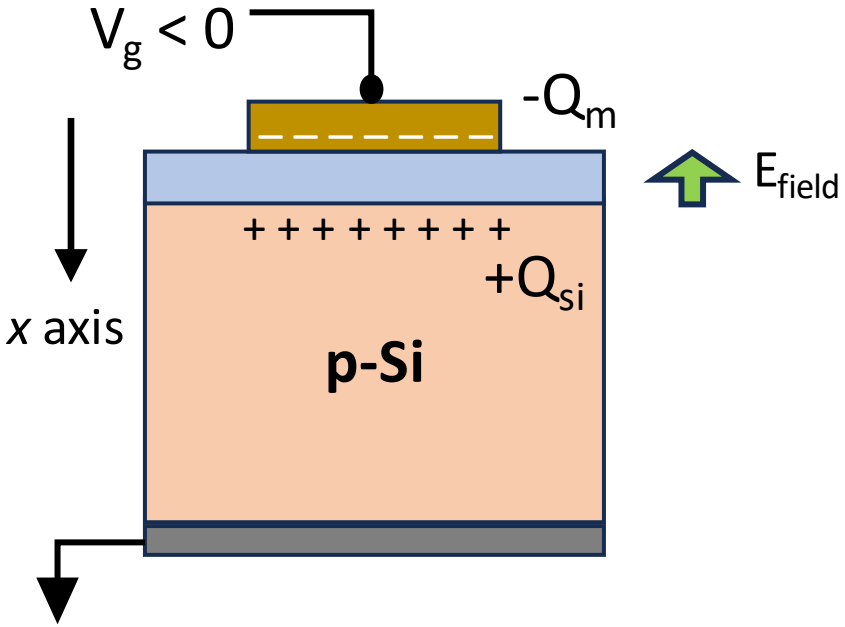

$V_g < 0$; $-Q_m$ on metal
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In *p*-Si: p+, n-, $N_A$- : Only p+

Let's take $N_A = 10^{15}$ cm$^{-3}$,
$p_0 = 10^{15}$ cm$^{-3}$; $n_0 = 10^5$ cm$^{-3}$



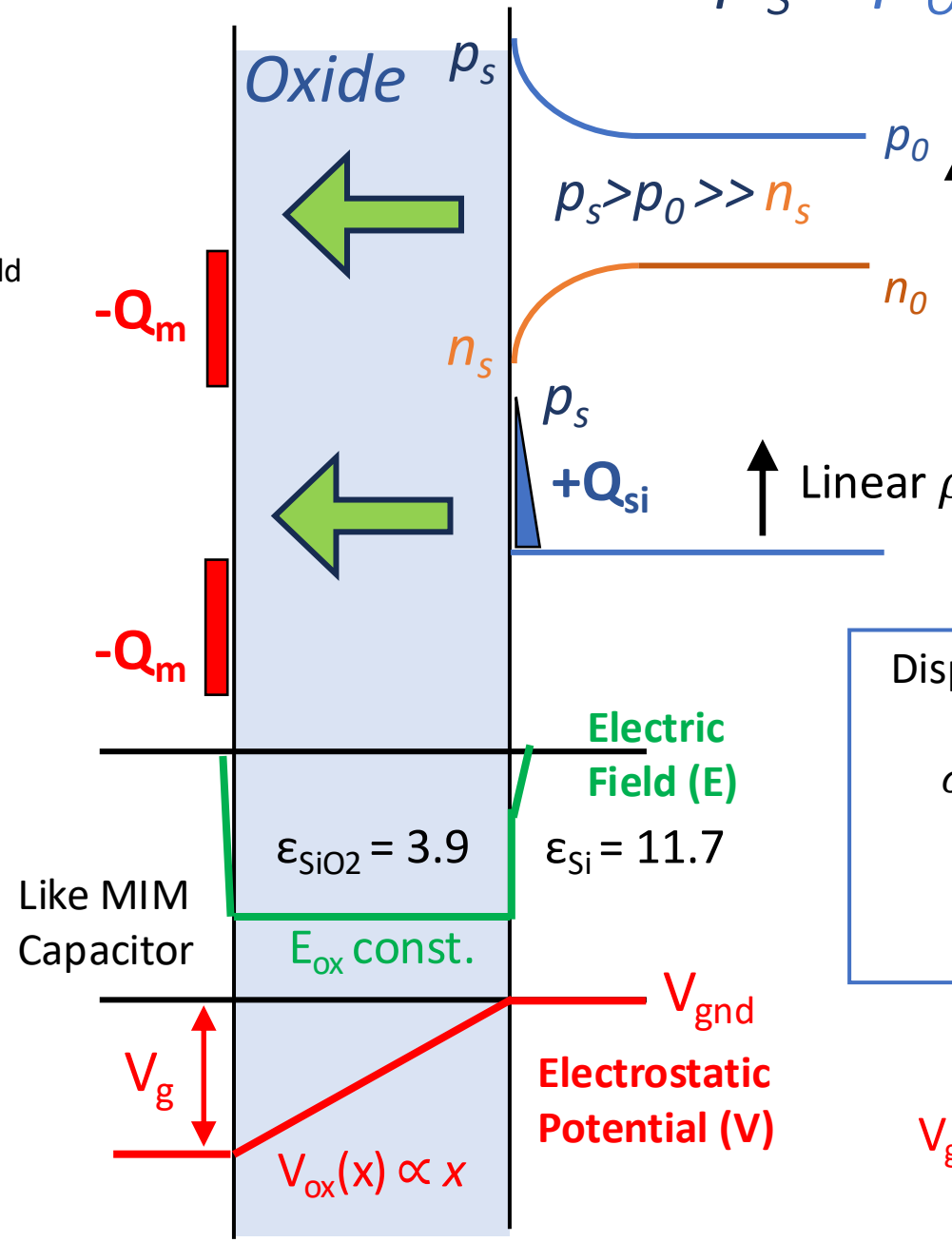

*Where $p_s$ Surface Accumulation of holes*

# $V_g < 0$ or Accumulation $p_s > p_0 >> n_s$




$V_g < 0$; $-Q_m$ on metal
Compensated by $+Q_{si} = -Q_m$
In *p*-Si: p+, n-, $N_A$- : Only p+

Let's take $N_A = 10^{15}$ cm$^{-3}$,
$p_0 = 10^{15}$ cm$^{-3}$; $n_0 = 10^5$ cm$^{-3}$




$$E(x) = \frac{1}{\varepsilon}\int \rho\,(x)\,dx$$

$$V(x) = -\int E\,(x)\,dx$$

***HW Q: Relationship b/w $E_g$ and $\varepsilon$, Why?***

Displacement (D = εE) continuity:

$$D_{ox} - D_{si} = \sigma_s$$

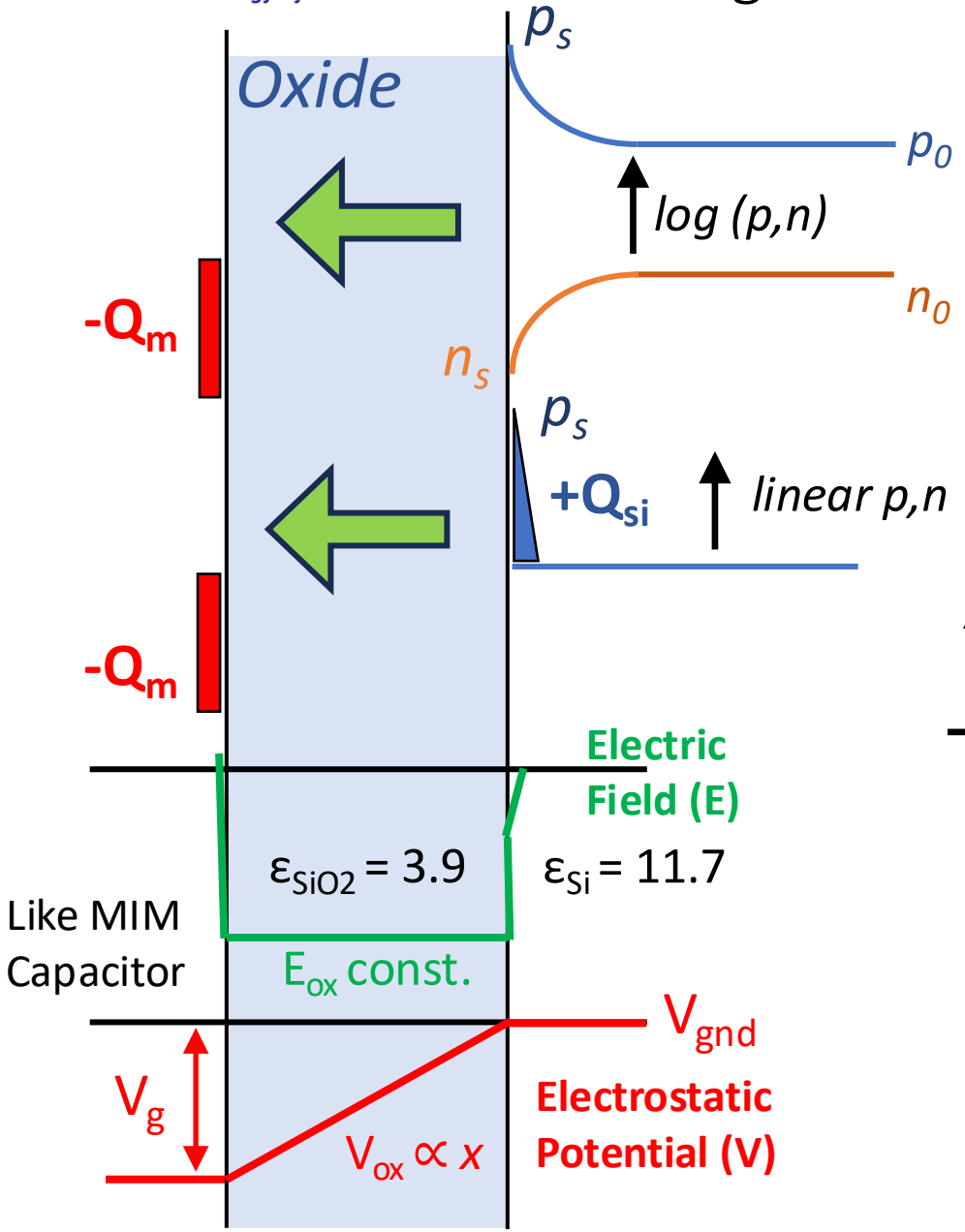
$\sigma_s$ -> surface charge density

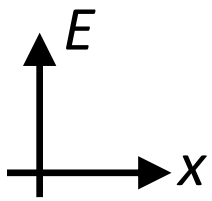
$$\varepsilon_{ox}E_{ox} - \varepsilon_{si}E_{si} = \sigma_s$$

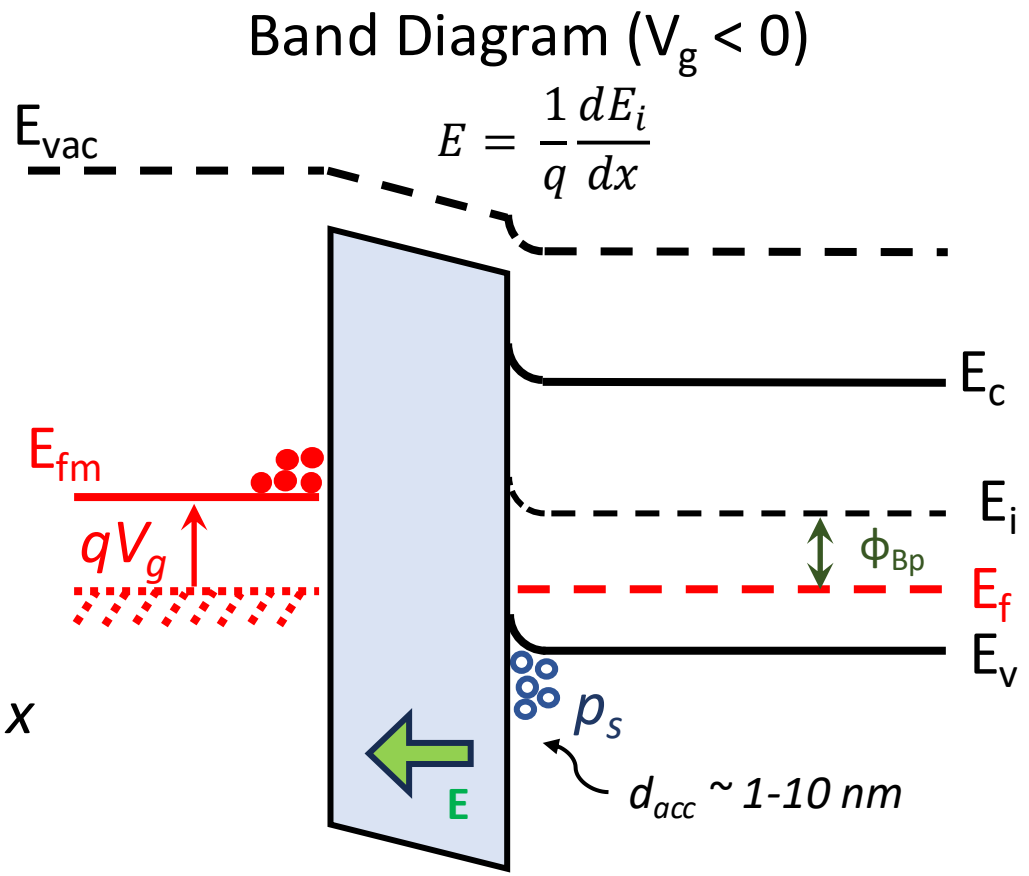
$$\sigma_s \approx 0 \rightarrow \frac{E_{ox}}{E_{si}} = \frac{\varepsilon_{si}}{\varepsilon_{ox}} \approx 3$$

$d_{acc}$ = 1-10 nm
$V_g \approx V_{ox}$ (All V drop on Ox)

# $V_g < 0$ or Accumulation $p_s > p_0 >> n_s$



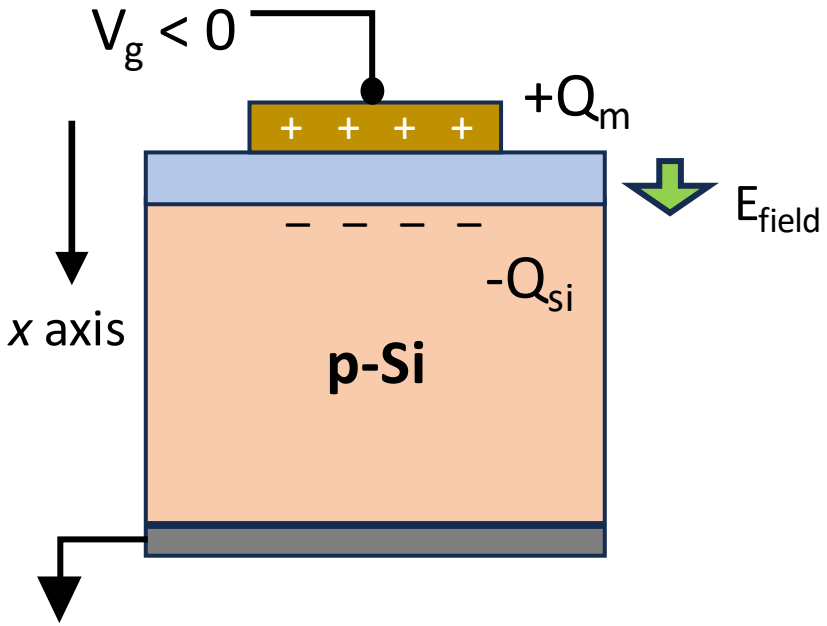




Important:

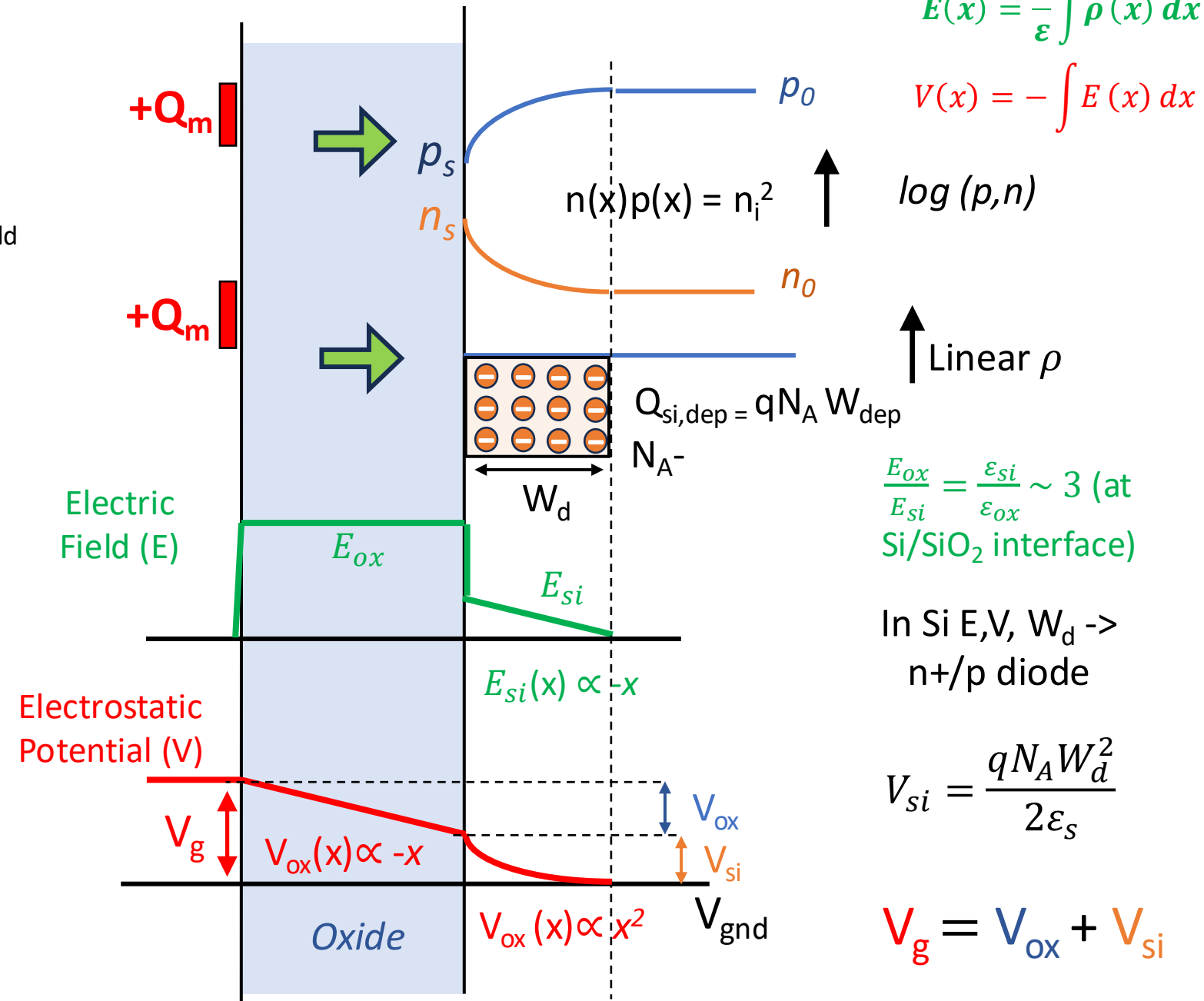
1) Metal; $E_{fm}$ moves up by $qV_g$
2) -ve gradient in $SiO_2$ bands
3) Sharp upwards bend in Si bands near Interface
   – $E_f$ constant; $E_{vac}$ bends with MOS bands

# $V_g > 0$ Depletion $p_0 > p_s > n_s$




$V_g > 0$; $+Q_m$ on metal
Compensated by $-Q_{si} = +Q_m$
In *p*-Si: p+, n-, $N_A$

Opposite to Accumulation,
*p* mobile charge moves away from (depletes) interface 'uncovering $N_A$-'
$-Q_{si} = -Q_{dep}$




$$E(x) = \frac{1}{\varepsilon}\int \rho(x)\,dx$$

$$V(x) = -\int E(x)\,dx$$

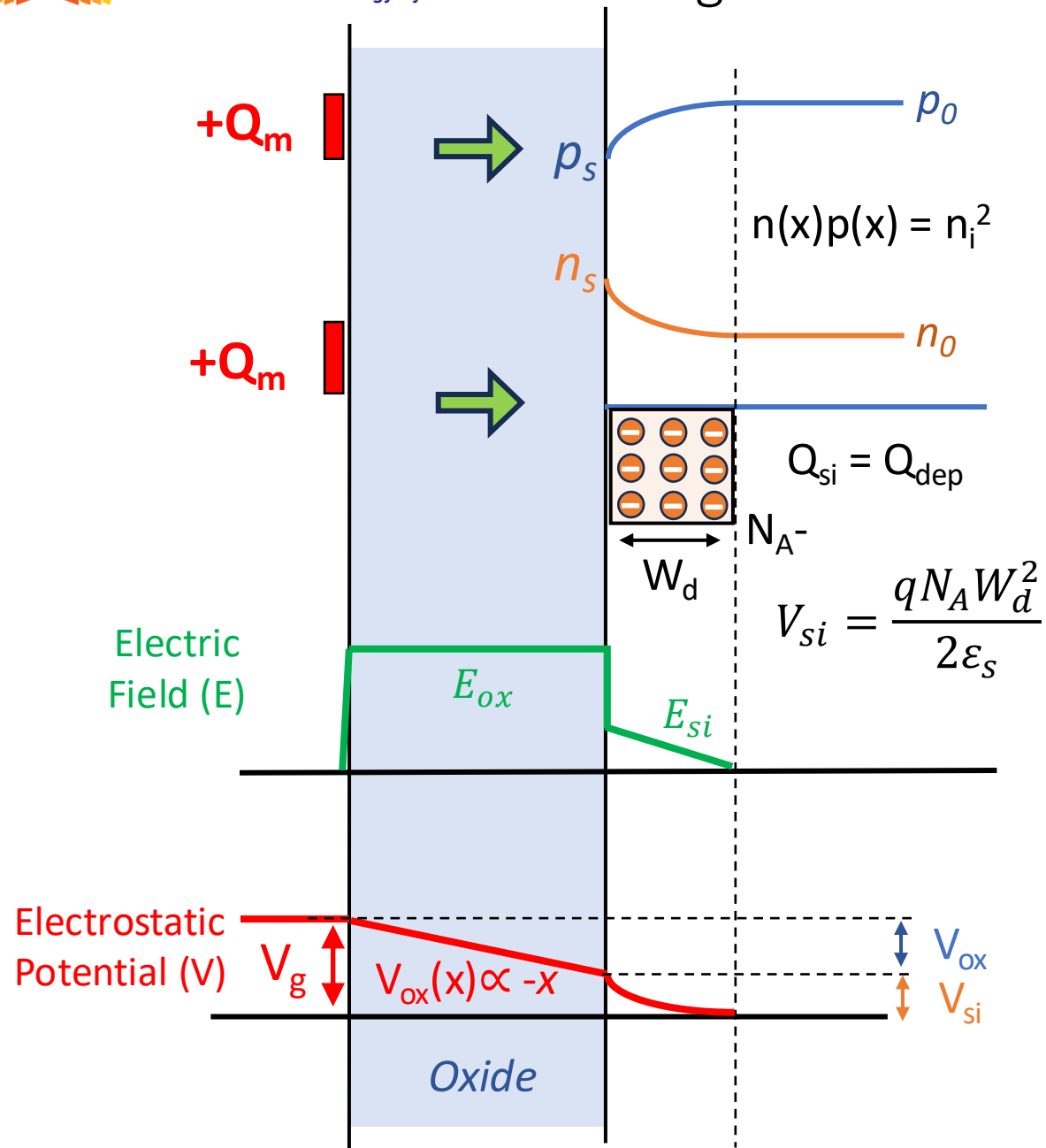
$$\frac{E_{ox}}{E_{si}} = \frac{\varepsilon_{si}}{\varepsilon_{ox}} \sim 3$$ (at $Si/SiO_2$ interface)

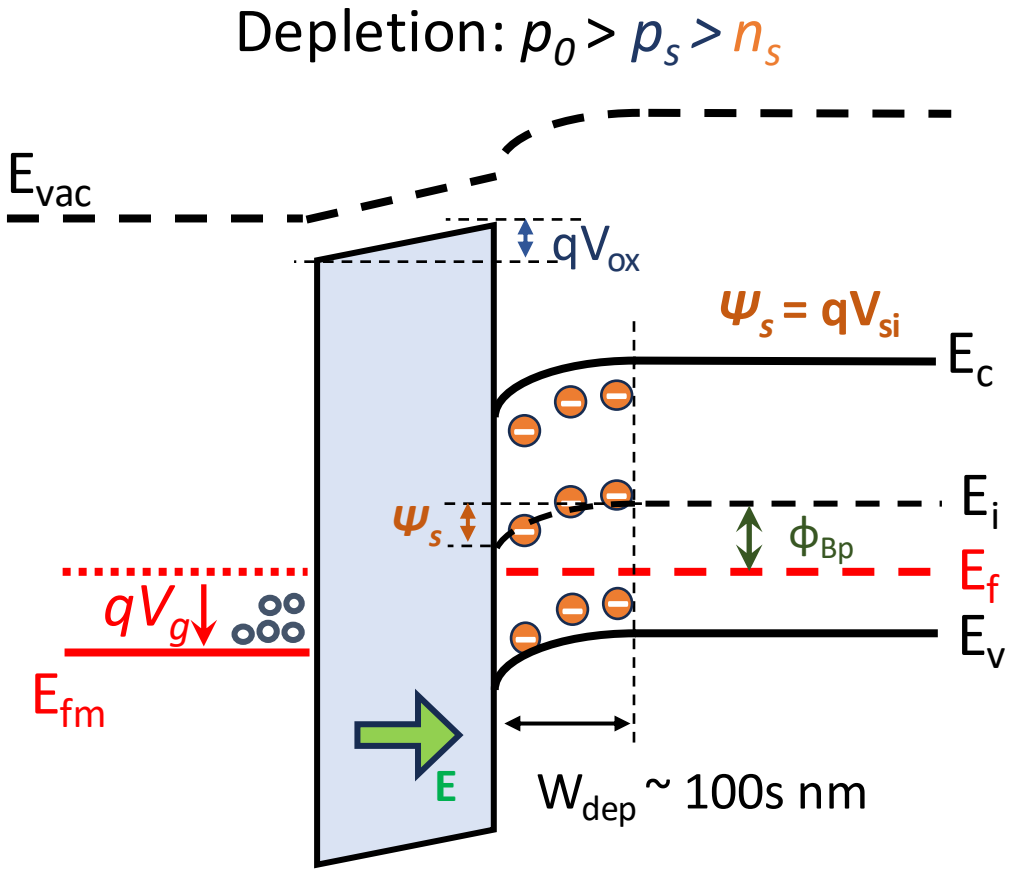
In Si E,V, $W_d$ -> n+/p diode

$$V_{si} = \frac{qN_A W_d^2}{2\varepsilon_s}$$

$$V_g = V_{ox} + V_{si}$$

# $V_g > 0$ (small) – Depletion $p_0 > p_s > n_s$

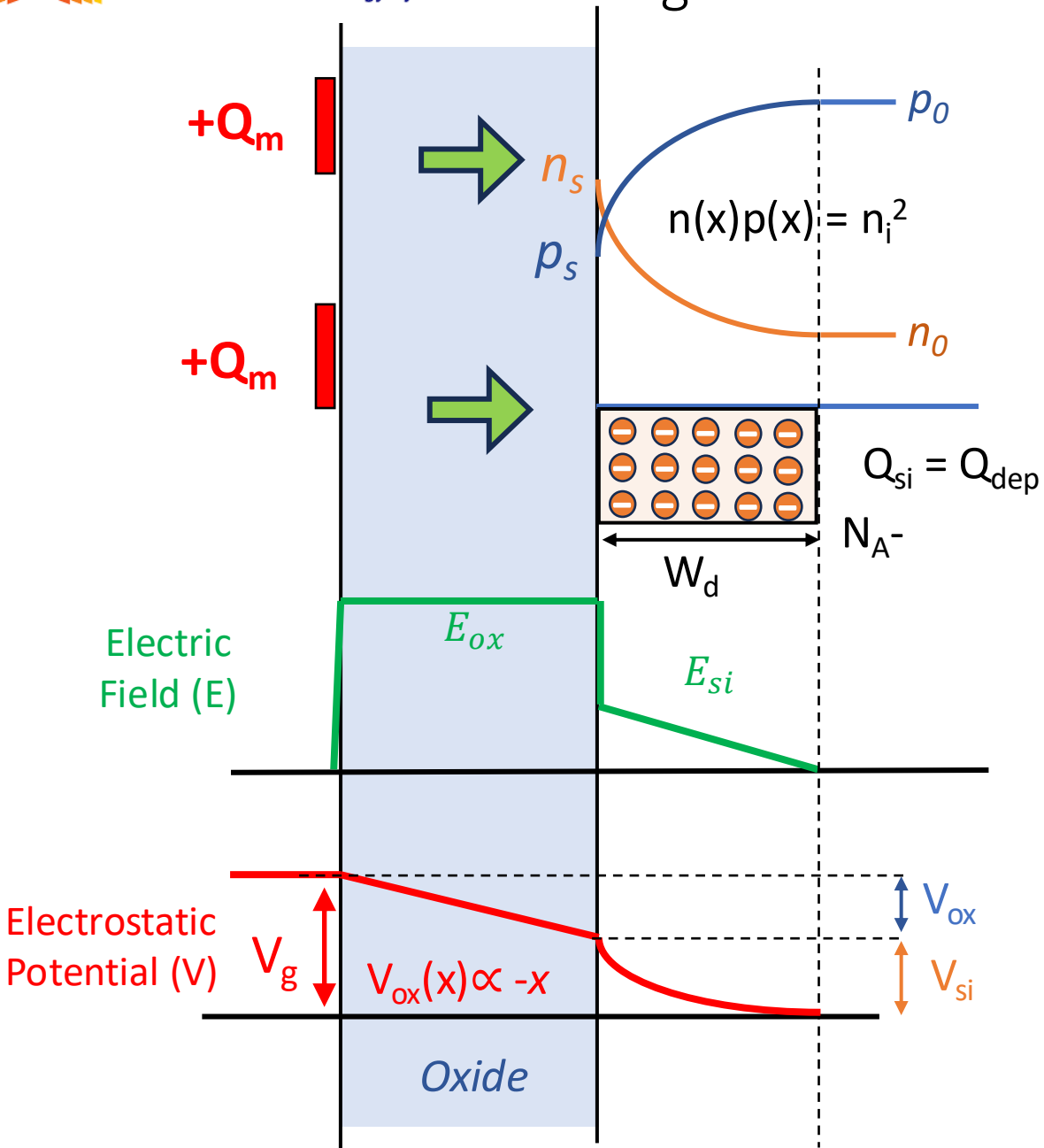



Important:

1) Metal; $E_{fm}$ moves down by $qV_g$
2) +ve gradient in $SiO_2$ bands
3) Gradual parabolic bend in Si bands @$W_{dep}$
4) $\psi_s = qV_{si}$ (Band bending @Si = Voltage drop)

# $V_g >> 0$ WEAK INVERSION $p_0 > n_s > p_s$

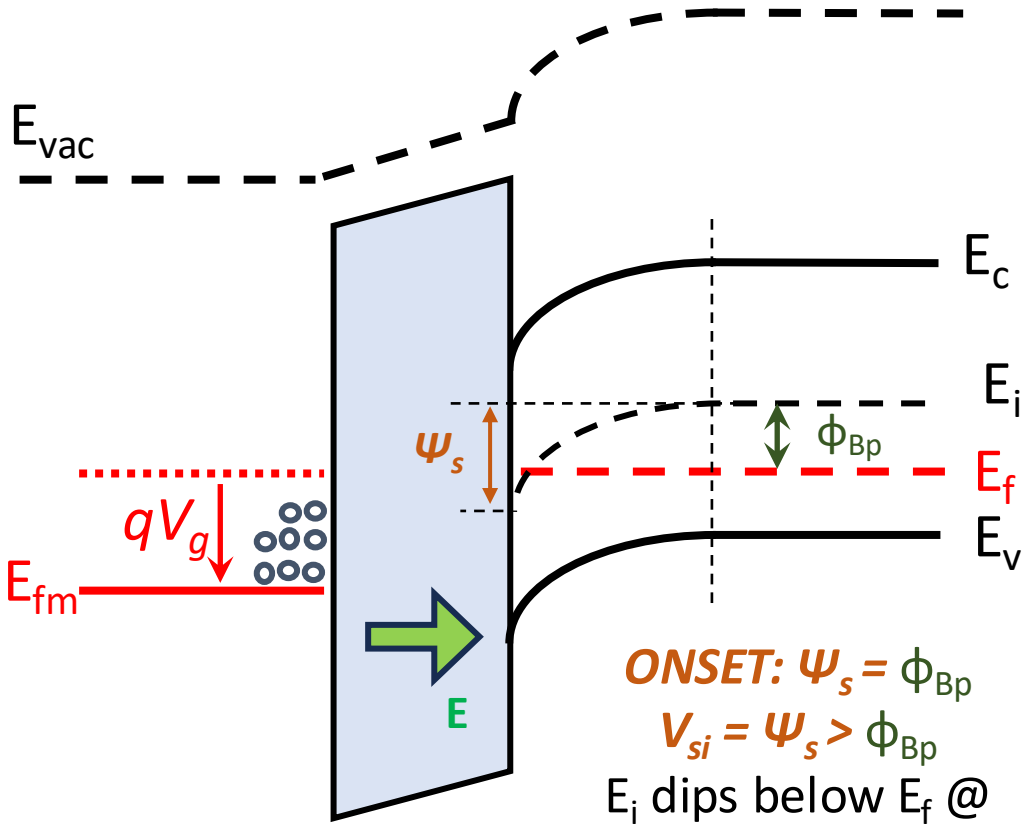



**WEAK INVERSION:**

Surface gets inverted from p->n

$p_0 > n_s > p_s$

*ONSET/START:* $n_s = p_s = n_i$

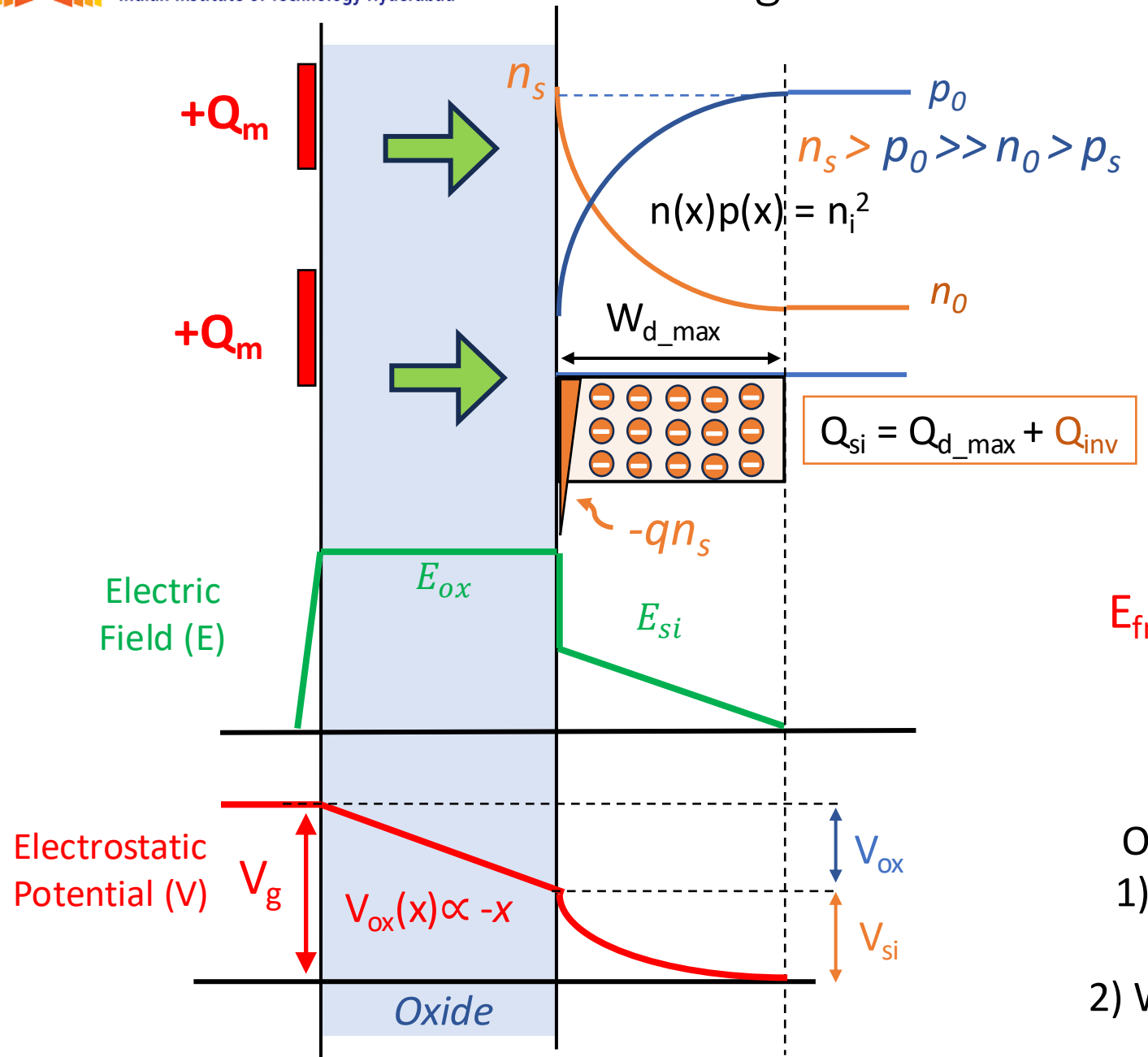



***ONSET:*** $\psi_s = \Phi_{Bp}$

$V_{si} = \psi_s > \Phi_{Bp}$

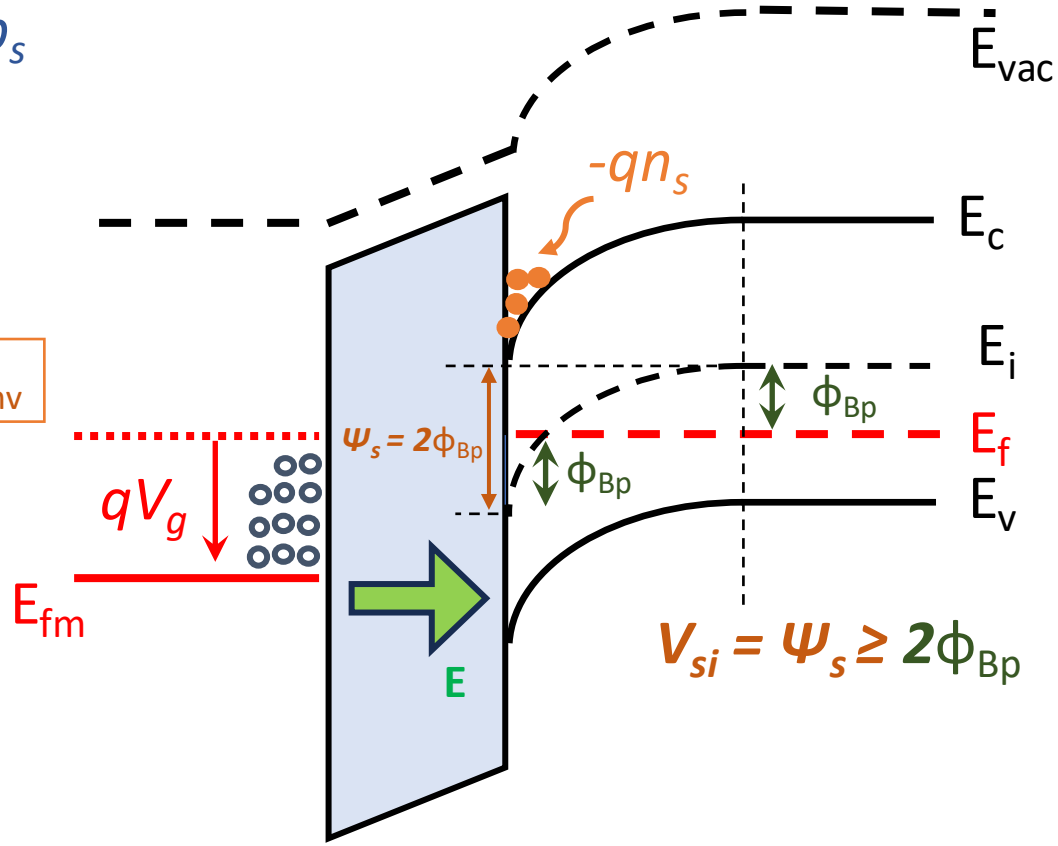
$E_i$ dips below $E_f$ @ Surface; $n_s > p_o$

# $V_g >> 0$ STRONG INVERSION $n_s > p_0$




**STRONG INVERSION:**

Surface gets completely inverted $n_s \geq p_0$




Onset of Strong Inversion: $n_s = p_0$ ; $\Psi_s = 2\phi_{Bp}$

1) $V_g = V_{th}$ (threshold voltage) -> formation of inversion channel ($-qn_s$) in MOSFETs

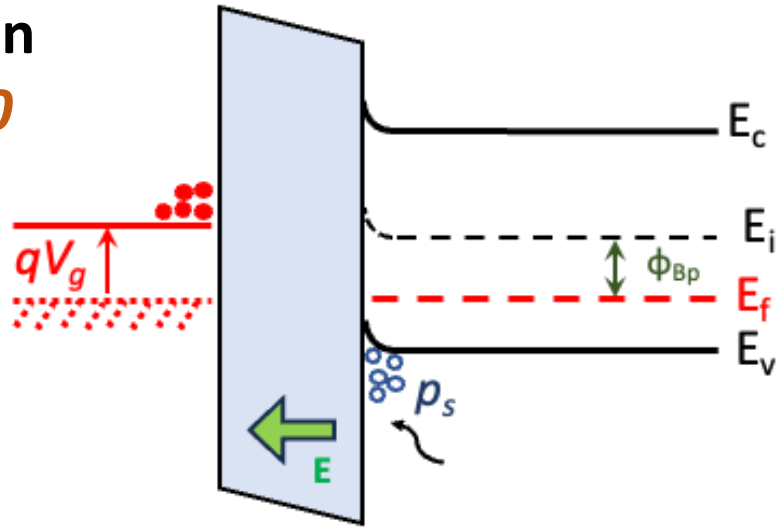
2) $W_{d_max}$ & $Q_{dmax}$ – Screening of $V_g$ due to ($-qn_s$)

# Recap MOSCAP Operation

## 1. Accumulation

$V_g < 0$ or $\psi_s < 0$

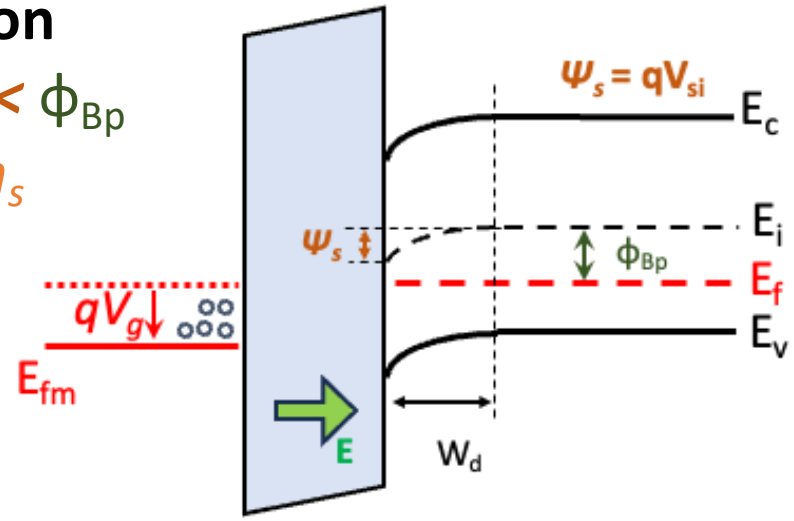
$p_s > p_0 >> n_s$




## 2. Depletion

$V_g > 0$; $0 < \psi_s < \phi_{Bp}$

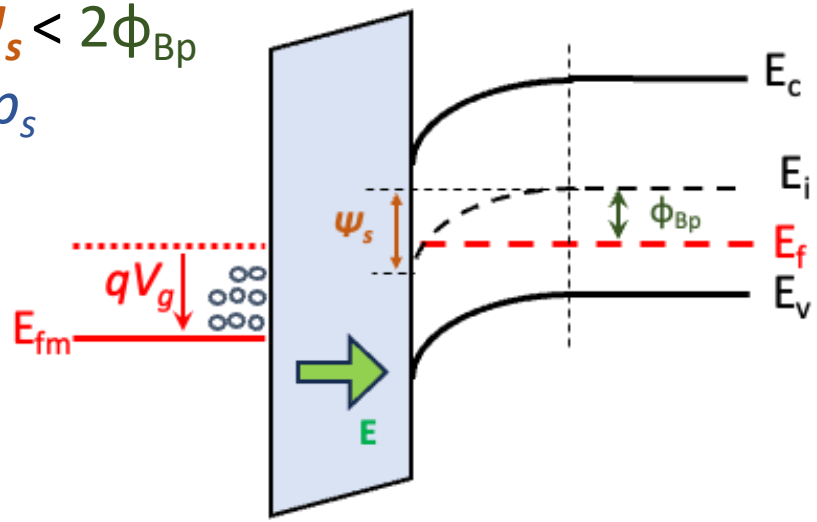
$p_0 > p_s > n_s$




## 3. Weak Inversion

$V_g >> 0$; $\phi_{Bp} < \psi_s < 2\phi_{Bp}$

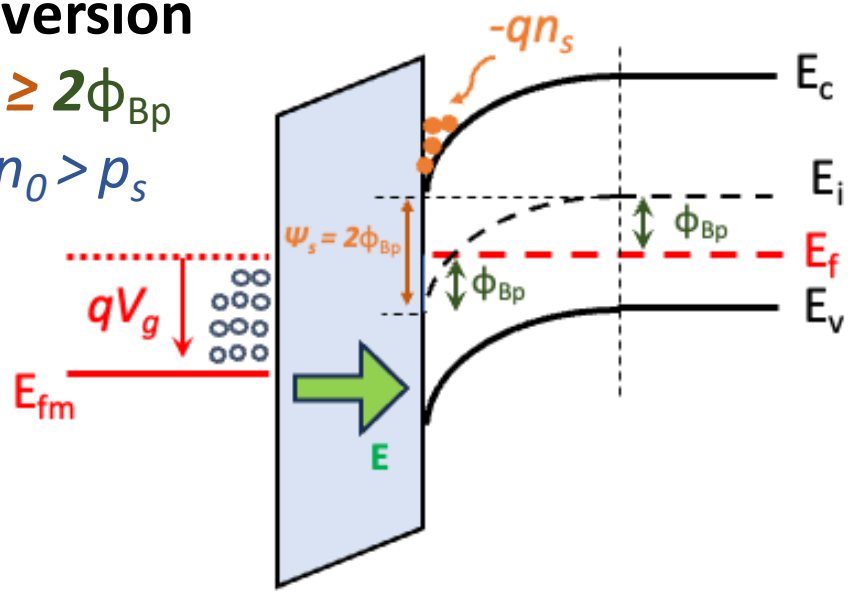
$p_0 > n_s > p_s$




## 4. Strong Inversion

$V_g >> 0$; $\psi_s \geq 2\phi_{Bp}$

$n_s > p_0 >> n_0 > p_s$

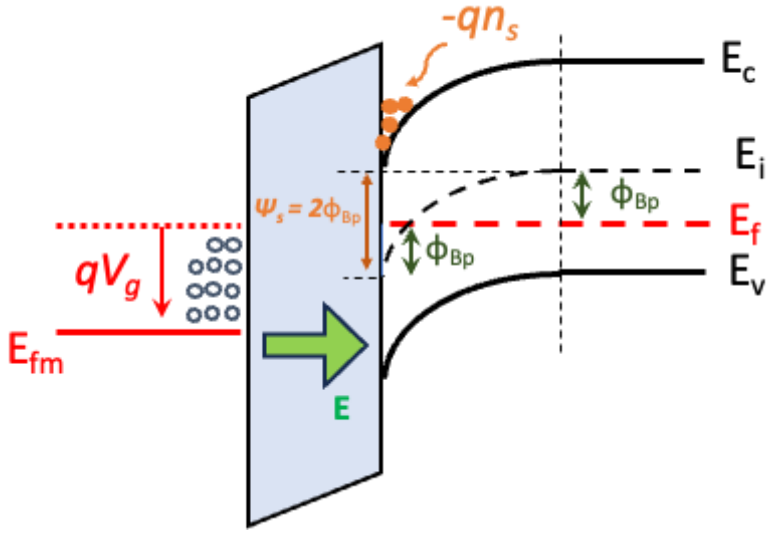


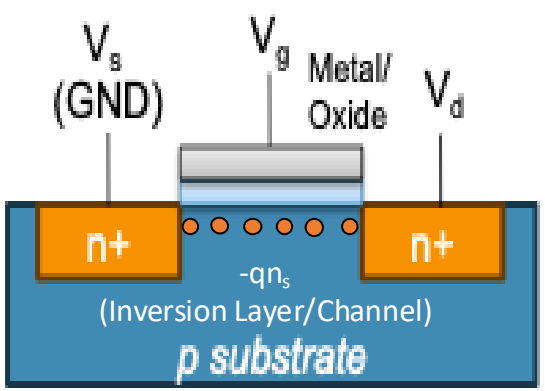
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

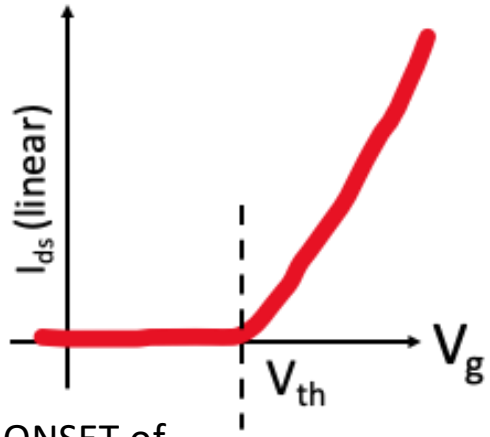
# Threshold Voltage ($V_{th}$) Concept & Derivation

Threshold Voltage Condition

$V_g = V_{th}$ when $\psi_s = 2\phi_{Bp}$




Importance of $V_{th}$ :ONSET of MOSFET channel formation, OFF -> ON transition

$$V_{th} = V_g[@\psi_s = 2\phi_{Bp}] = V_{si} + V_{ox} = \psi_s + \frac{Q_g = Q_{si}}{C_{ox}} *$$

**Onset of Strong Inversion** ($n_s = p_0$) at $V_g = V_{th}$

$$(a)\psi_s = 2\phi_{Bp} = 2V_t \ln\left(\frac{N_A}{n_i}\right); (b) W_d = W_{d,max}$$

$$V_{th} = 2\phi_{Bp} + \frac{Q_{si}}{C_{ox}};$$ How to calculate $Q_{si}$ ?

$$Q_{si} = Q_{d,max} + * Q_{inv} \approx Q_{d,max} = -qN_A W_{d,max}$$

Recall, one-sided junction: $W_d = \sqrt{\frac{2\varepsilon_s \psi_s}{qN_A}}$

Substituting $\psi_s = 2\phi_{Bp}$: $W_{d,max} = \sqrt{\frac{2\varepsilon_s(2\phi_{Bp})}{qN_A}} = \sqrt{\frac{4\varepsilon_s \phi_{Bp}}{qN_A}}$

Substituting $W_{d,max}$ in $Q_{si} \approx Q_{d,max} = -\sqrt{4q\varepsilon_s N_A \phi_{BP}}$

Substituting $Q_{si}$ in $V_{th}$

$$V_{th} = 2\phi_{Bp} + \frac{\sqrt{4q\varepsilon_s N_A \phi_{Bp}}}{C_{ox}}$$

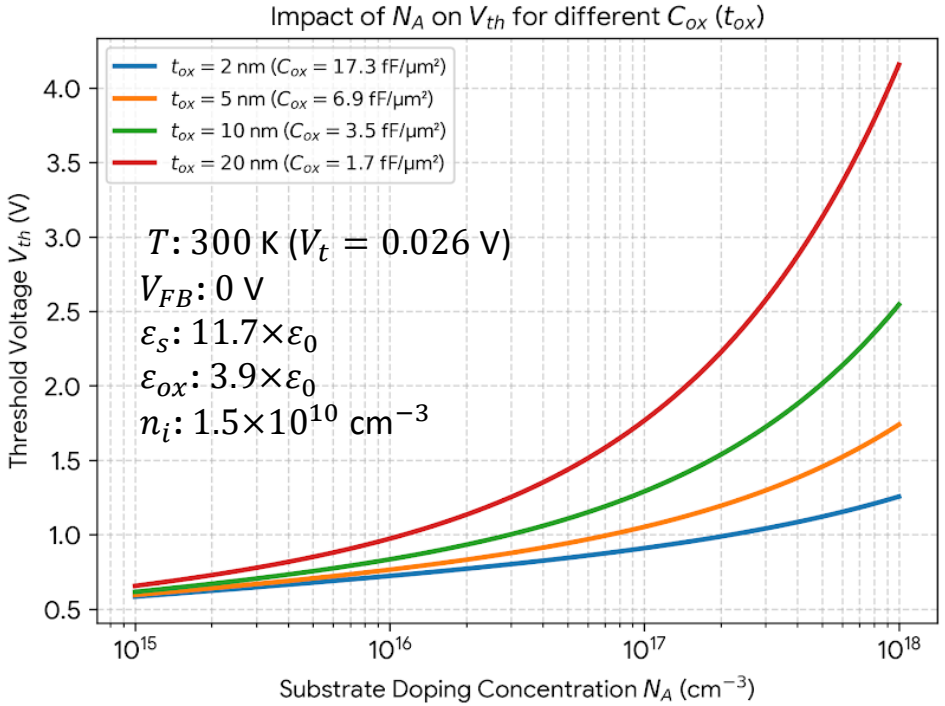
***Note 1:** $Q_{si}$ and $C_{ox}$ are Charge and Capacitance per unit gate area (Area gets cancelled, therefore the convention)*

*Note 2: *Why $Q_{inv}$ was ignored, Because $d_{inv}$ (1-3 nm) << $W_d$ (100's nm)*

# Threshold Voltage ($V_{th}$) Concept & Derivation

$$V_{th} = 2\phi_{Bp} + \frac{\sqrt{4q\varepsilon_s N_A \phi_{Bp}}}{C_{ox}};$$

$$where\ \phi_{Bp} = V_t \ln\left(\frac{N_A}{n_i}\right)\ \&\ * C_{ox} = \frac{\varepsilon_{ox}}{t_{ox}}$$



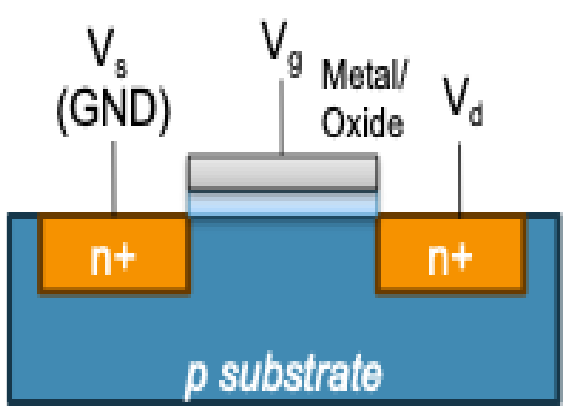

$N_A$ ⇧ $C_{ox}$ ⇩ -> $V_{th}$ ⇧

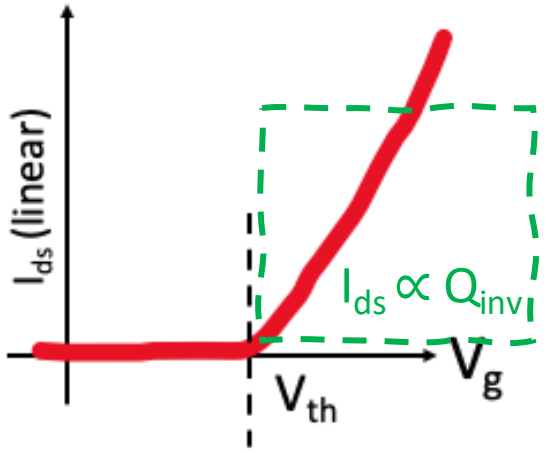
Explain why - Physically!

Question: If $W_d$ stops growing $V_g > V_{th}$
Where does the charge go?

Answer: Inversion layer! $Q_{si} = Q_{dep}$ (fixed) $+ Q_{inv}$

$\Delta Q_{si} = \Delta Q_{inv} = C_{ox}(V_{gs} - V_{th})$ -> parallel plate capacitor ($C_{ox}$)



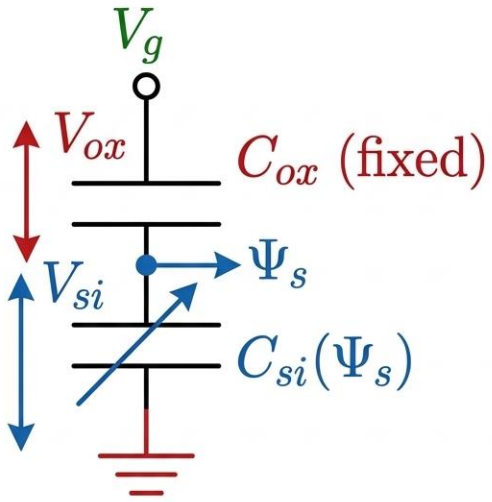




Since $I_{ds} \propto Q_{mobile,\ Si}$

For $V_{gs} > V_{th}$ -> $I_{ds} \propto Q_{inv} = C_{ox}(V_{gs} - V_{th})$

Ideal MOSFET $I_{ds}$ linear with $V_{gs}$ for $V_{gs} > V_{th}$

# ~~Deriving~~ Arriving at CV Characteristics




$$\frac{1}{C_g} = \frac{1}{C_{ox}} + \frac{1}{C_{si}}$$

Smaller of $(C_{ox}, C_{si}) \approx C_g$

$$C_{ox} \equiv \text{fixed} = \frac{\epsilon_{ox}}{t_{ox}}$$

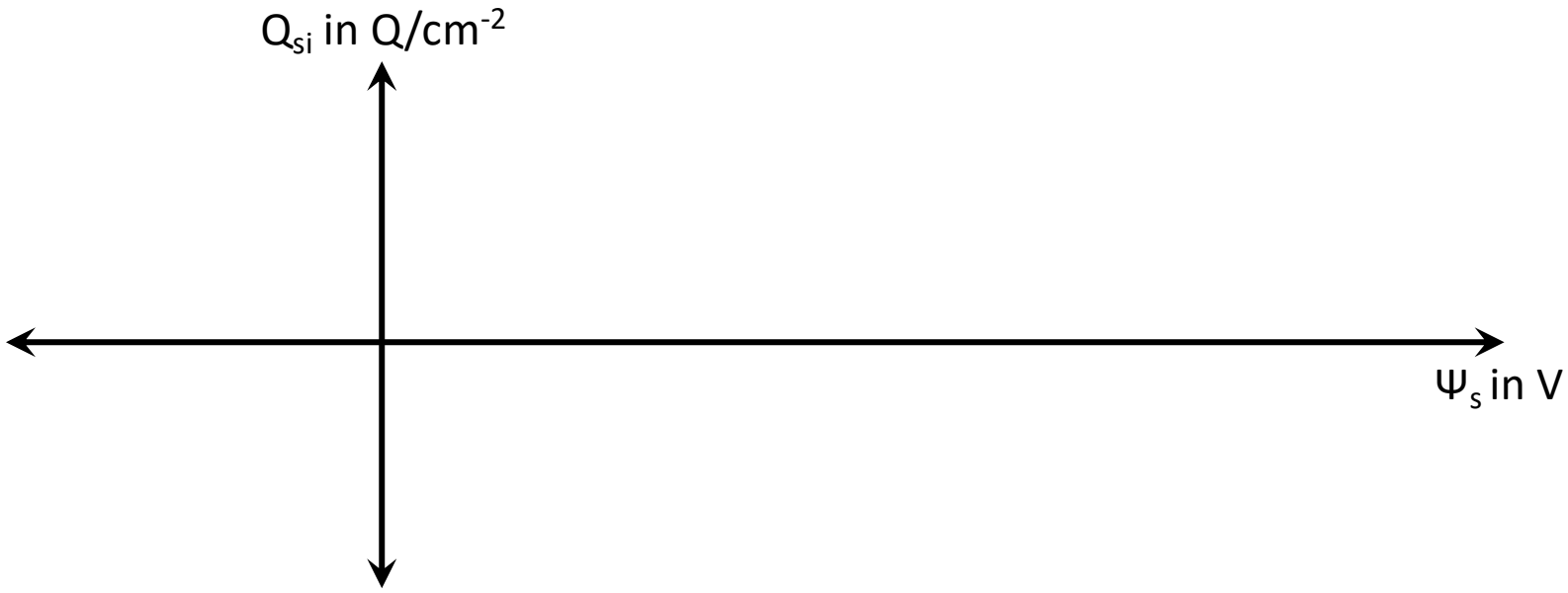
$$C_{si} = \frac{dQ_{si}}{dV_{si}} = \frac{dQ_{si}}{d\Psi_s}$$

{Small signal}

{Changes with Operation conditions}

{Recall $C_{ox}, C_{si}$ per unit area}

Instead of doing a formal derivation,
Plot $Q_{si}$ vs $\Psi_s$ intuitively in the 4 modes of operation



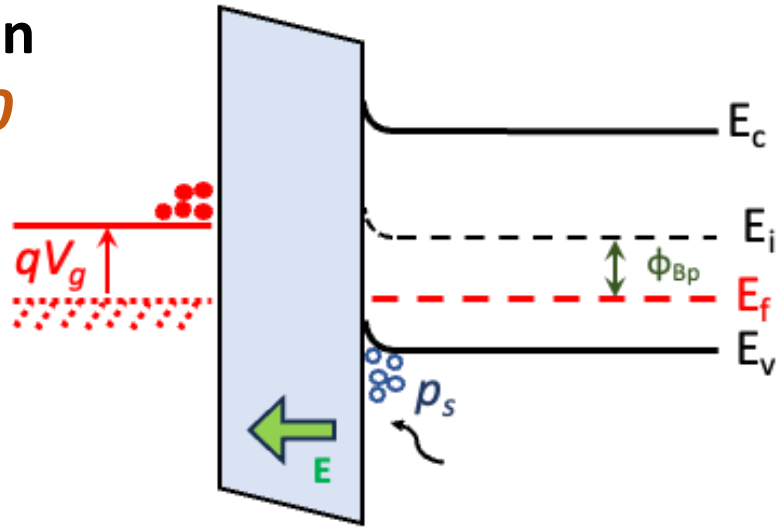

Step 1: Mark on 4 modes & Source of $Q_{si}$

# Recap MOSCAP Operation

## 1. Accumulation

$V_g < 0$ or $\psi_s < 0$

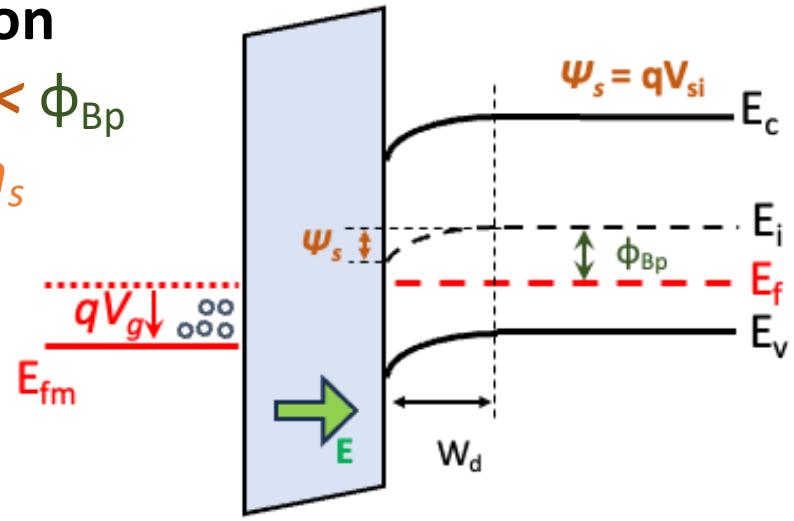
$p_s > p_0 >> n_s$




## 2. Depletion

$V_g > 0$; $0 < \psi_s < \phi_{Bp}$

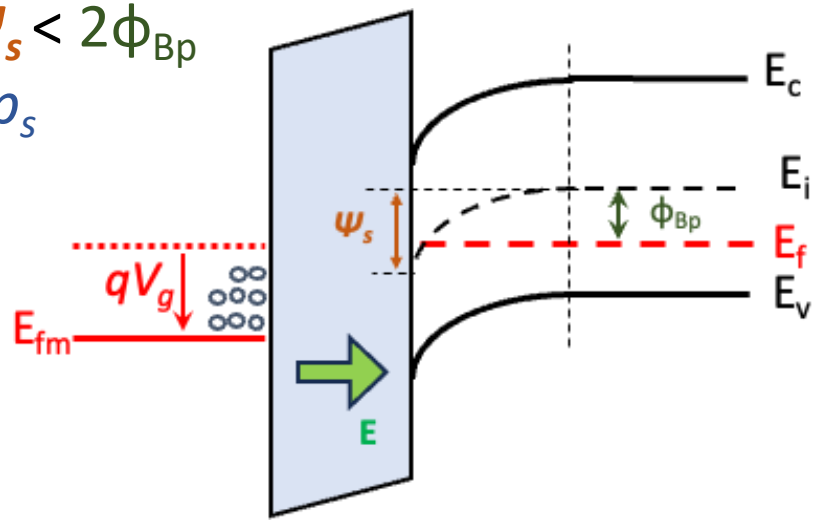
$p_0 > p_s > n_s$




## 3. Weak Inversion

$V_g >> 0$; $\phi_{Bp} < \psi_s < 2\phi_{Bp}$

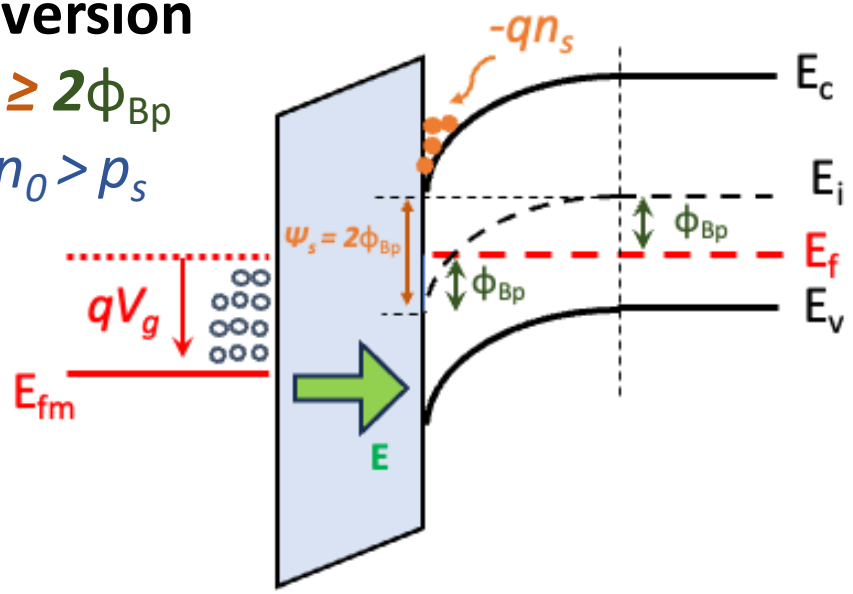
$p_0 > n_s > p_s$



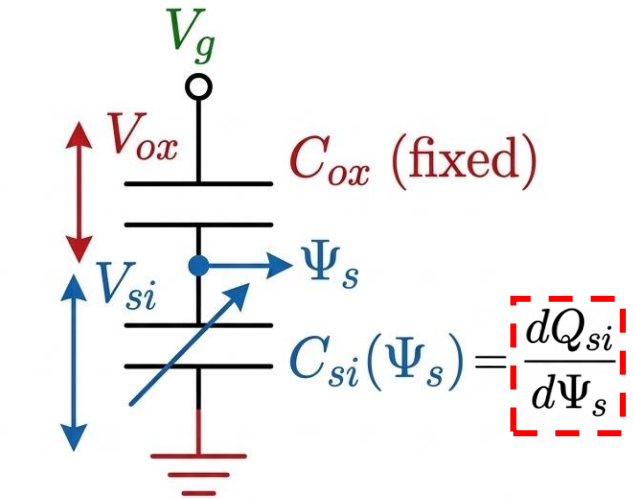

## 4. Strong Inversion

$V_g >> 0$; $\psi_s \geq 2\phi_{Bp}$

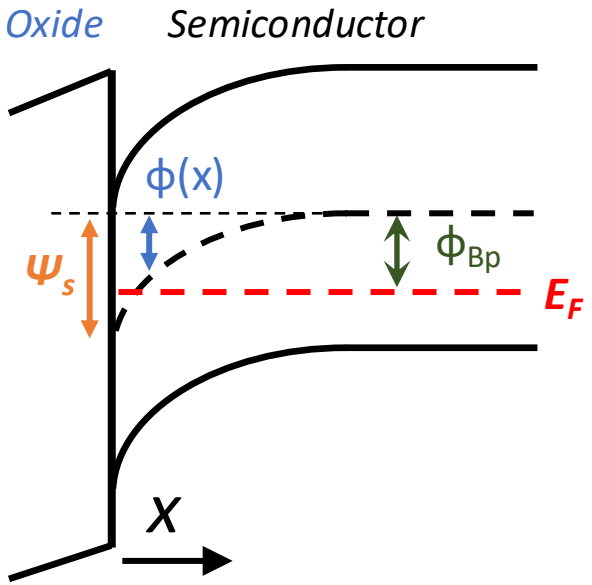
$n_s > p_0 >> n_0 > p_s$



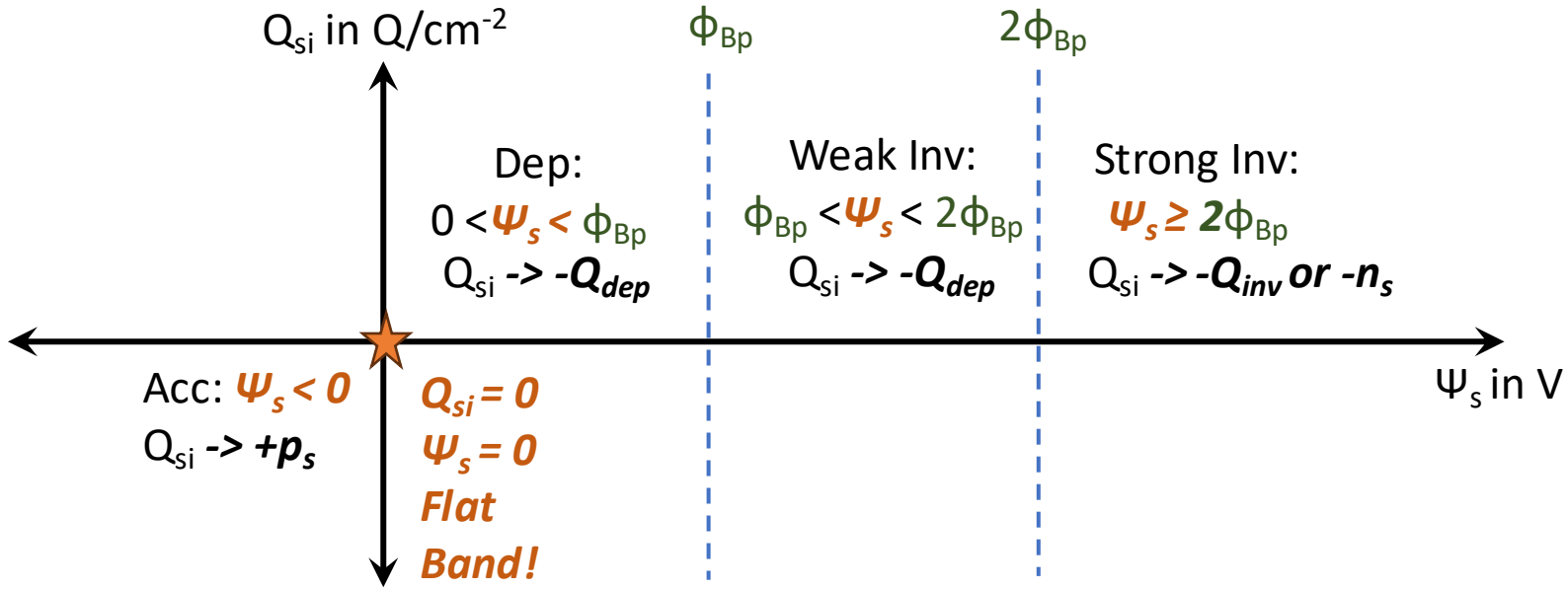
# ~~Deriving~~ Arriving at CV Characteristics




$$\frac{1}{C_g} = \frac{1}{C_{ox}} + \frac{1}{C_{si}}$$

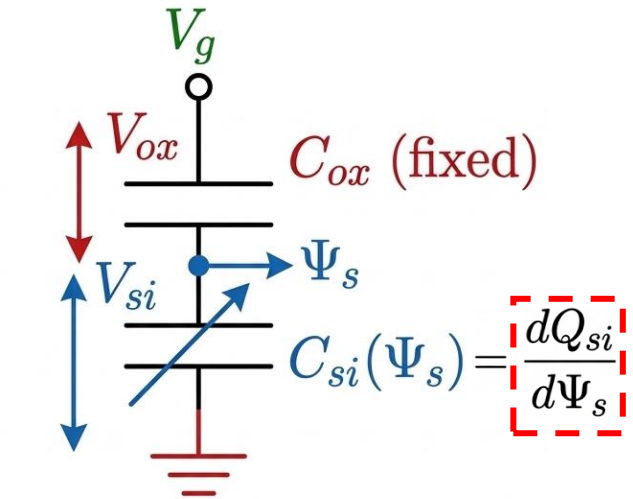



Instead of doing a formal derivation,
Plot $Q_{si}$ vs $\Psi_s$ intuitively in the 4 modes of operation

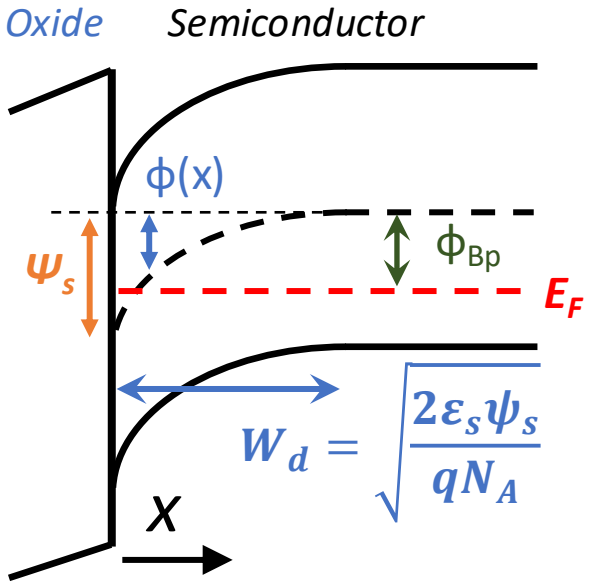



Step 1: Mark on 4 modes & Source of $Q_{si}$

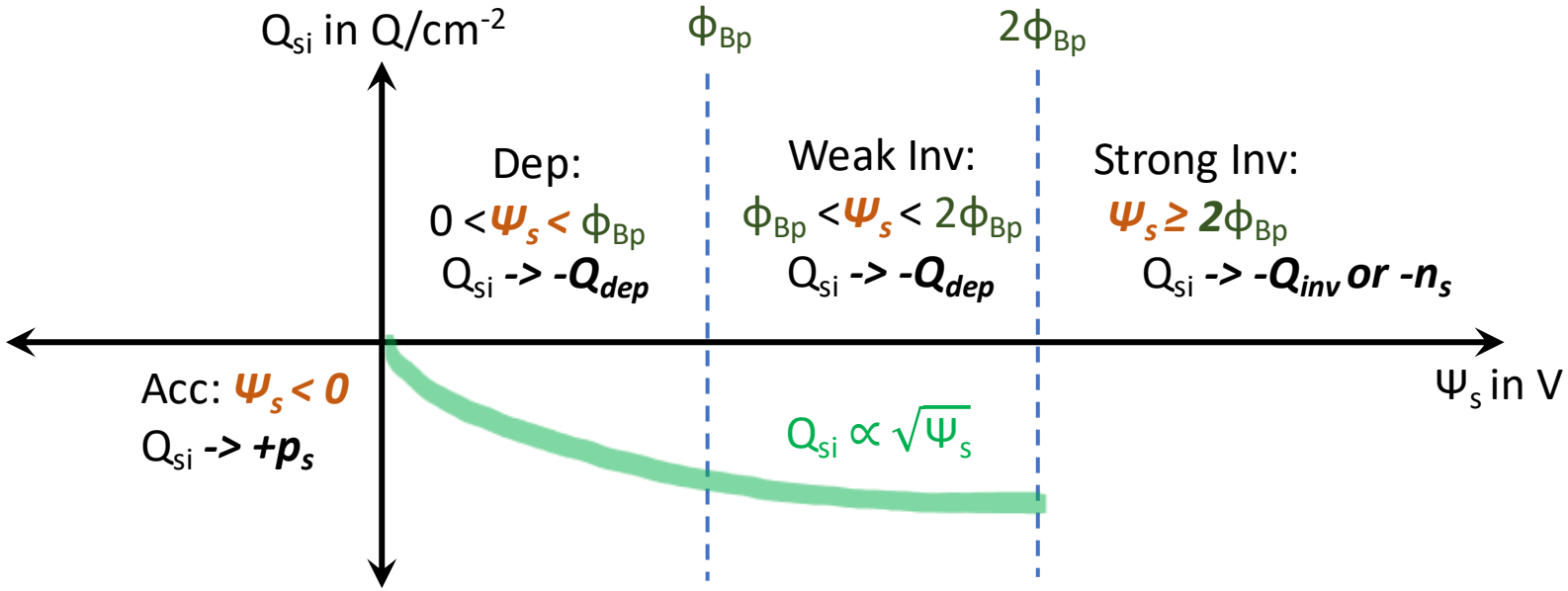
# ~~Deriving~~ Arriving at CV Characteristics




$$\frac{1}{C_g} = \frac{1}{C_{ox}} + \frac{1}{C_{si}}$$




Instead of doing a formal derivation,
Plot $Q_{si}$ vs $\Psi_s$ intuitively, the 4 modes of operation

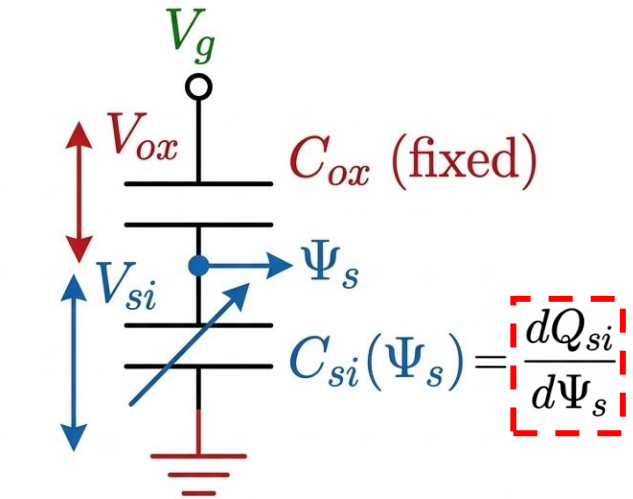



Step 2: Estimate functional dependence of $Q_{si}$ with $\Psi_s$

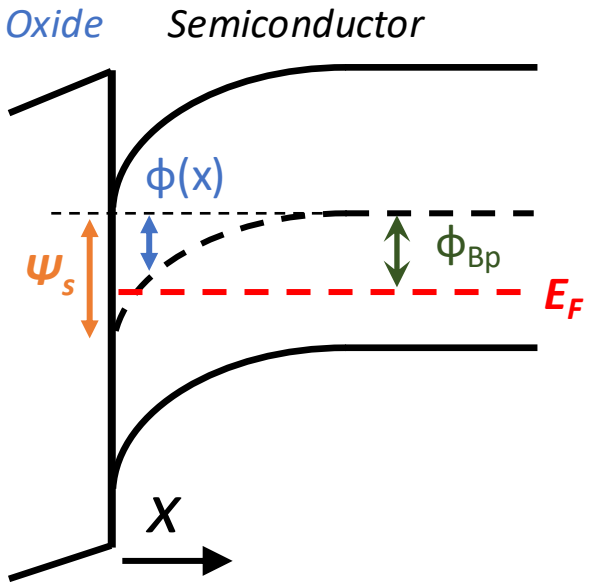
$V_{fb} = 0$; $Q_{si} = 0$ @ $\Psi_s$ $V_g = 0$

Depletion and Weak Inv: $Q_{si} = -Q_{dep} = -qN_AW_d \propto \sqrt{\Psi_s}$

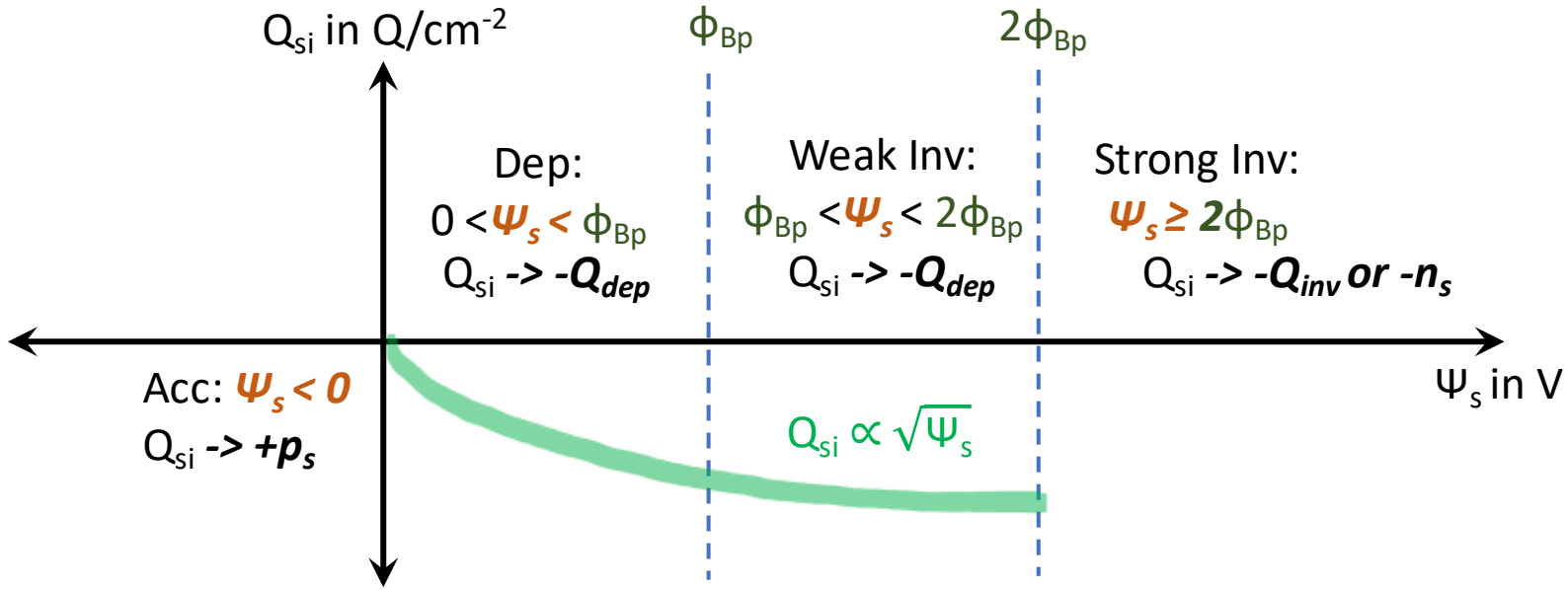
# ~~Deriving~~ Arriving at CV Characteristics




$$\frac{1}{C_g} = \frac{1}{C_{ox}} + \frac{1}{C_{si}}$$

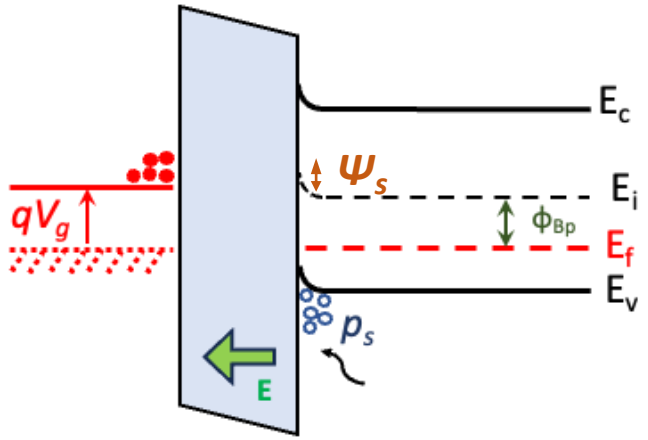



Instead of doing a formal derivation,
Plot $Q_{si}$ vs $\Psi_s$ intuitively, the 4 modes of operation

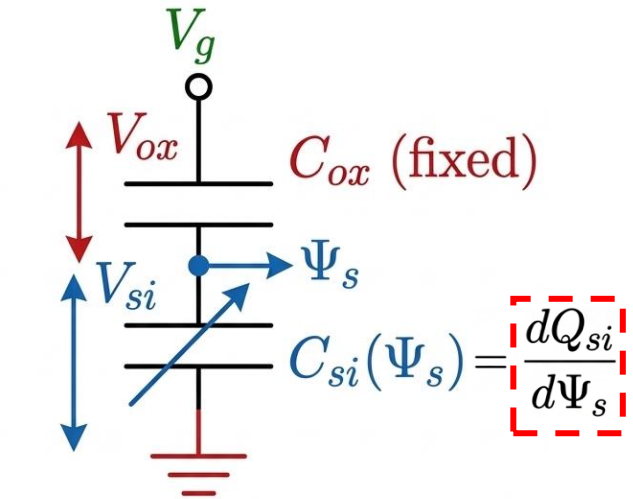



Step 2: For Acc & Strong Inv:
Let's review Band Diagrams

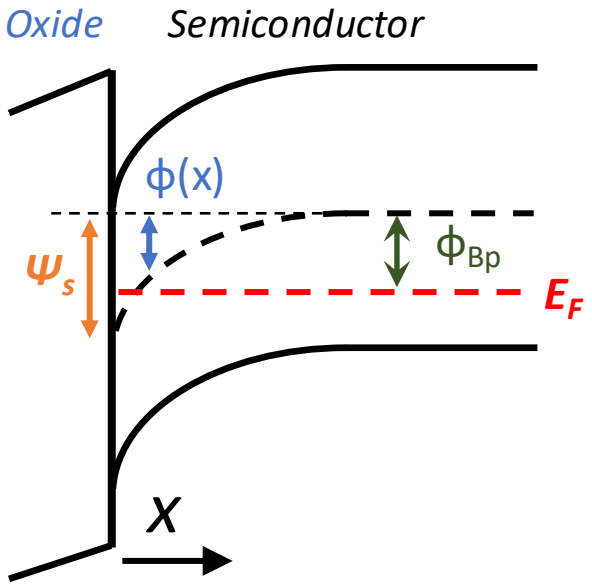
$p_s \propto \exp(-(N_{vs}-E_f) = \psi_s)$
$n_s \propto \exp(-(N_{cs}-E_f) = \psi_s)$
Therefore, $Q_{si} \propto \exp(\psi_s)$



# ~~Deriving~~ Arriving at CV Characteristics

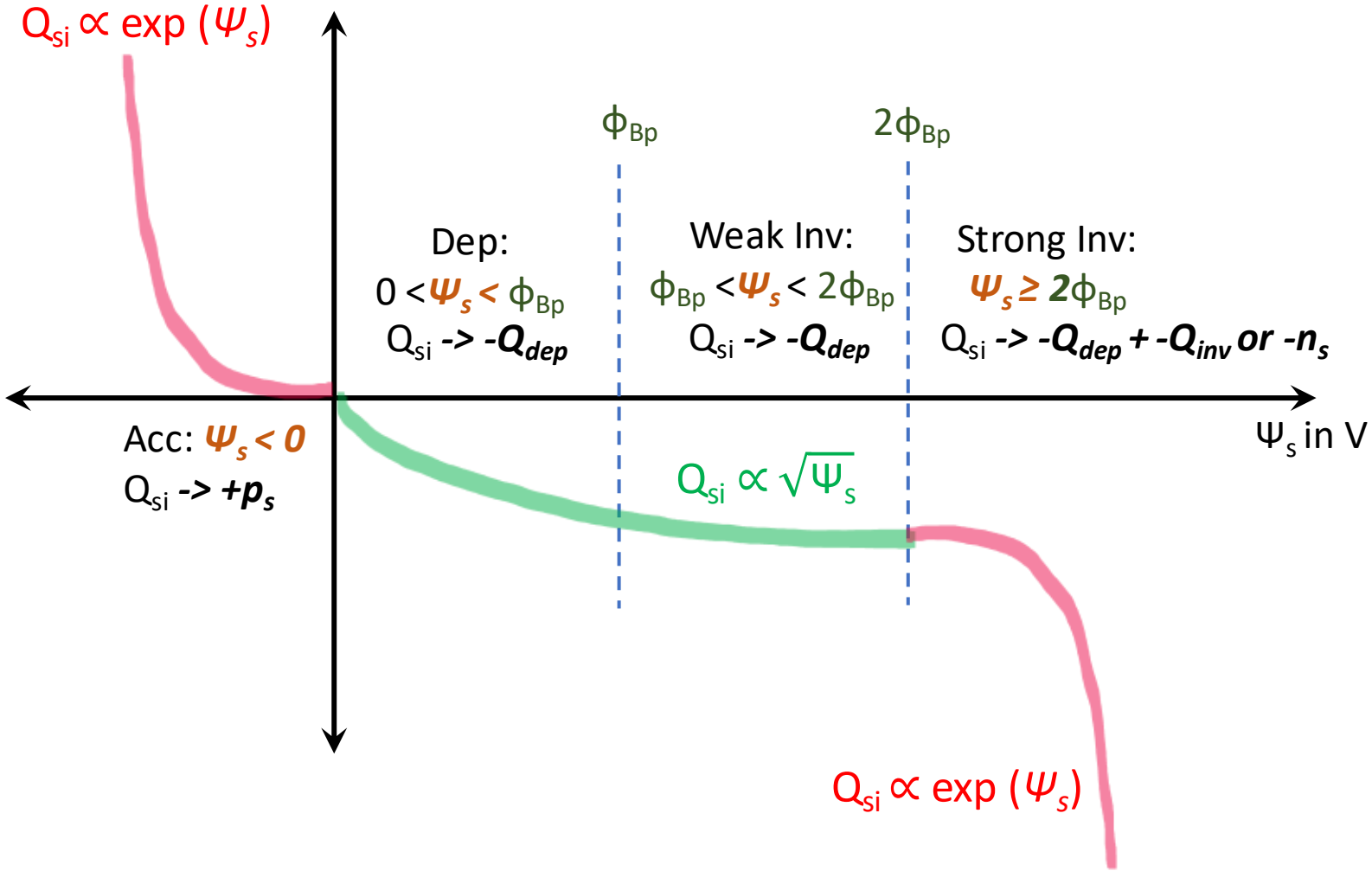



$$\frac{1}{C_g} = \frac{1}{C_{ox}} + \frac{1}{C_{si}}$$

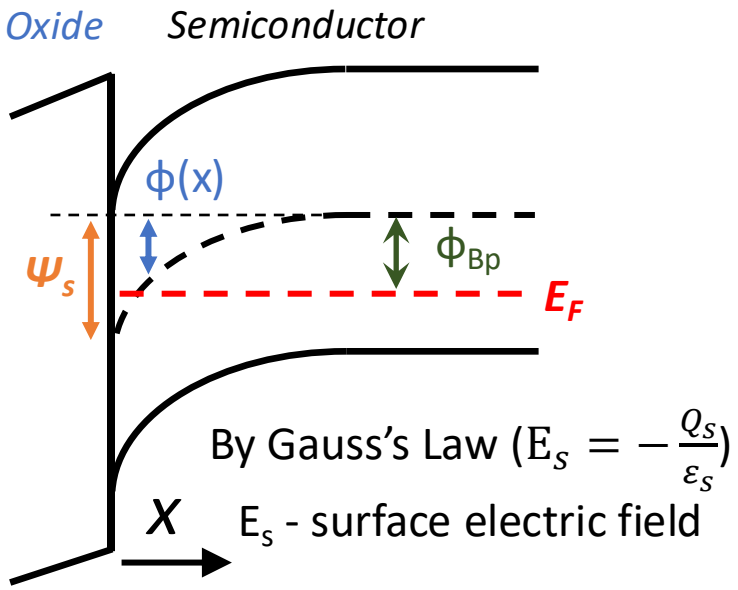



Instead of doing a formal derivation,
Plot $Q_{si}$ vs $\Psi_s$ intuitively, the 4 modes of operation

$Q_{si}$ in Q/cm$^{-2}$



# Deriving $Q_{si} - \Psi_s$ – matches our intuition

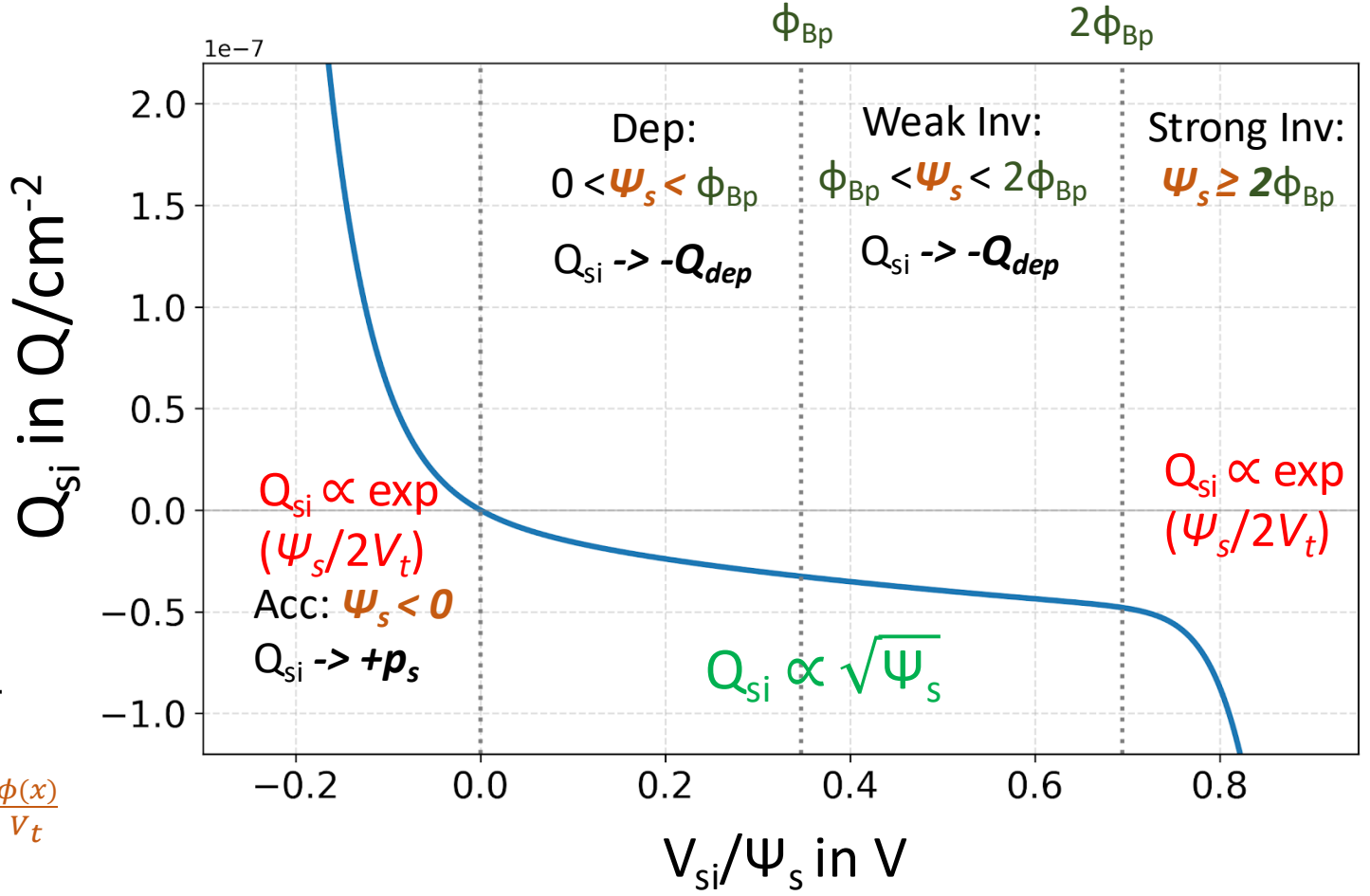



By Gauss's Law ($E_s = -\frac{Q_s}{\varepsilon_s}$)

$E_s$ - surface electric field

$$\frac{d^2\phi(x)}{dx^2} = -\frac{\rho(x)}{\varepsilon_s} = -\frac{q[p(x)-n(x)-N_A]}{\varepsilon_s}$$

$$p(x) = N_A e^{-\frac{\phi(x)}{V_t}} \text{ and } n(x) = n_{p0} e^{\frac{\phi(x)}{V_t}} = \left(\frac{n_i^2}{N_A}\right) e^{\frac{\phi(x)}{V_t}}$$

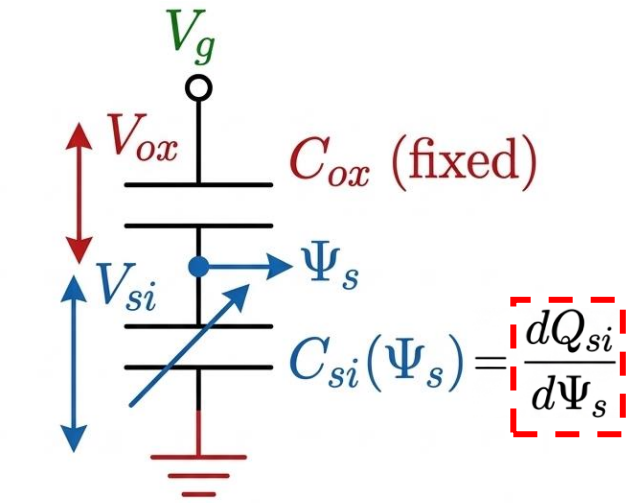



Complete Derivation in
Solid State Electronic Devices (7th Edition)
Streetman – Banerjee - Section 6.4.2
(Not important for this course)

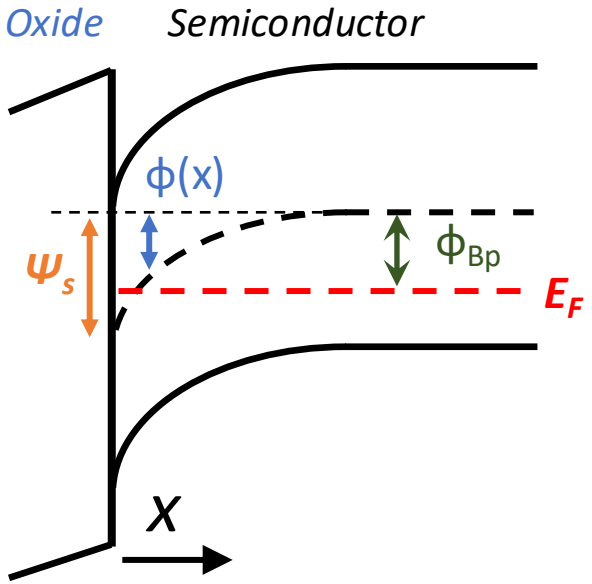
Solving of Poisson's Equation gives qualitatively same results as our intuitive reasoning!

*Important: $Q_{si} \propto \exp(\Psi_s/2V_t)$ Pre-factor of ½ in Acc & Strong Inversion*

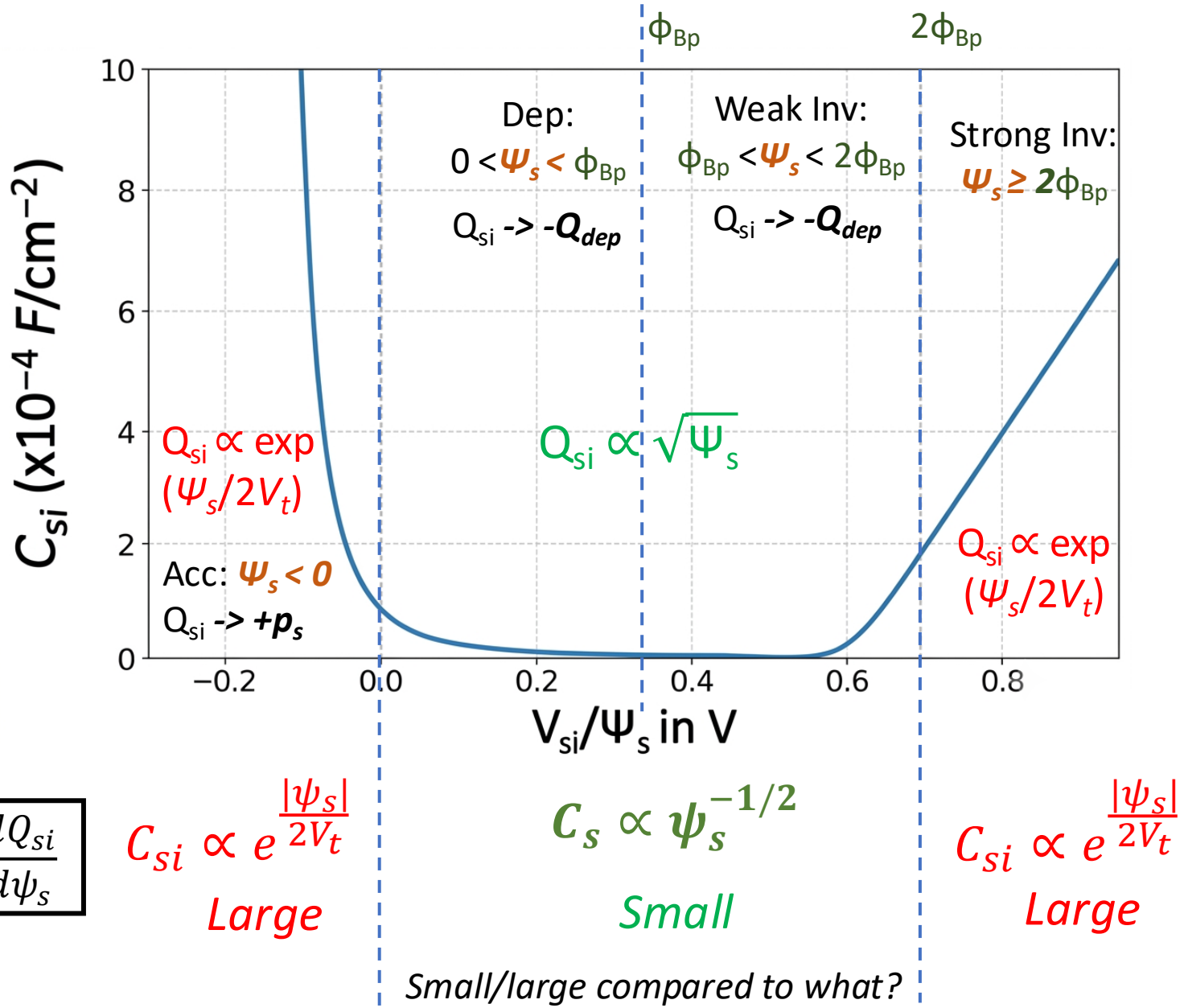
# Differentiating $Q_{si}(\psi_s)$ to get $C_{si}$



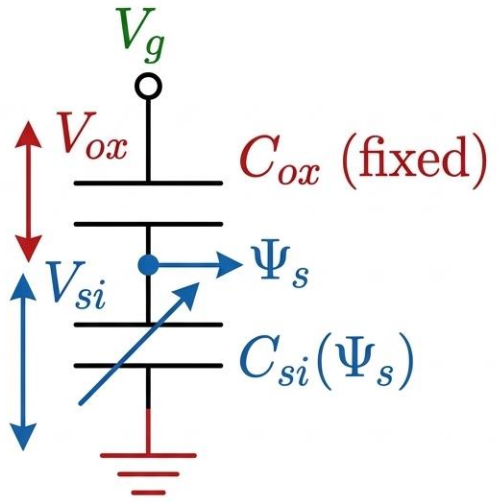

$$\frac{1}{C_g} = \frac{1}{C_{ox}} + \frac{1}{C_{si}}$$




$$C_{si} = \frac{dQ_{si}}{d\psi_s}$$



# Arriving at $C_g(\psi_s)$ $C_{si}$ in series with $C_{ox}$




$$\frac{1}{C_g} = \frac{1}{C_{ox}} + \frac{1}{C_{si}}$$

Smaller of $(C_{ox}, C_{si}) \approx C_g$

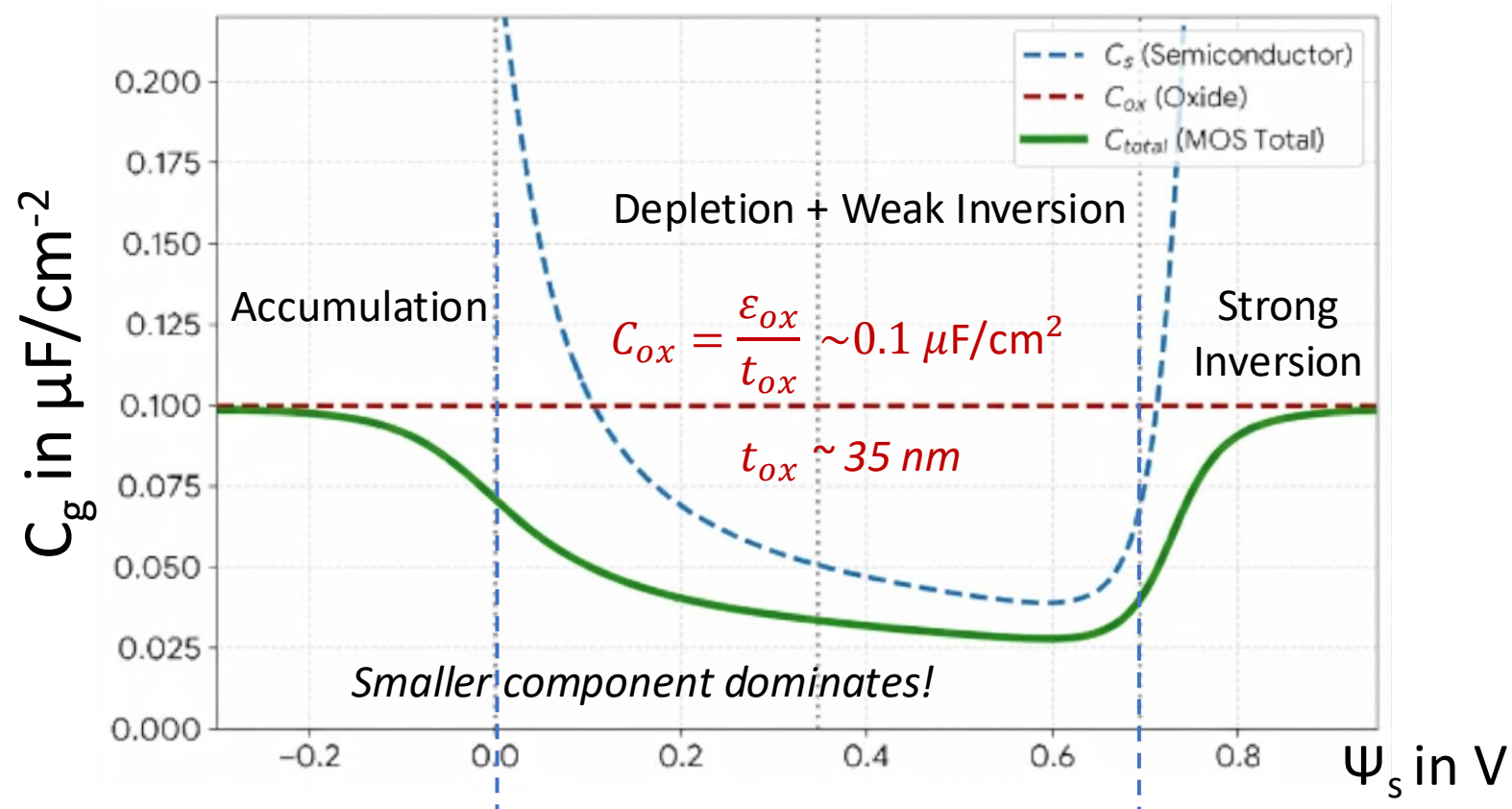
$$C_{ox} \equiv \text{fixed} = \frac{\epsilon_{ox}}{t_{ox}}$$

$$C_{si} = \frac{dQ_{si}}{dV_{si}} = \frac{dQ_{si}}{d\Psi_s}$$

{Small signal}

{Changes with Operation conditions}

{Recall $C_{ox}, C_{si}$ per unit area}




$C_{g,total} \approx C_{ox}$

$C_g$ as if a || Cap With $C_{ox}$

$$C_{g,total} = \frac{C_{ox}C_{dep}}{C_{ox} + C_{dep}}$$

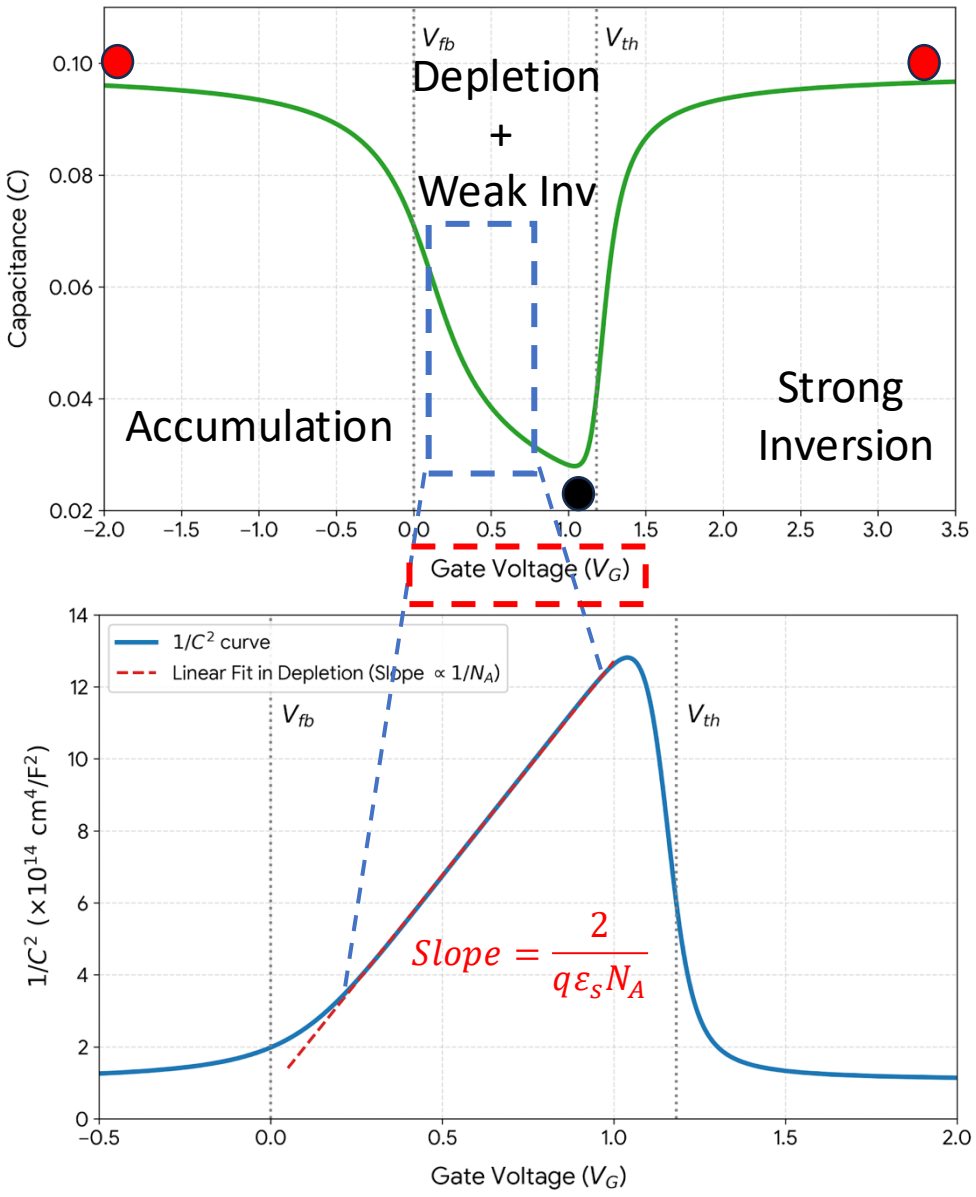
$$C_{dep} = \frac{\varepsilon_{si}}{W_{si}}$$

$C_g$ -> $C_{ox}$ series $C_{dep}$

$C_{g,total} \approx C_{ox}$

$C_g$ as if a || Cap With $C_{ox}$

# MOSCAP CV & extraction of parameters




Transforming $\Psi_s$ to $V_g$ (Voltage divider) $\quad \frac{dV_G}{d\psi_s} = \frac{C_{ox}+C_{si}}{C_{ox}} = m$ *(Body Factor)*

- 1) $C_{g,max} \approx C_{ox}$ at Strong Inversion and Accumulation; $t_{ox} = \varepsilon_{ox}/C_{ox}$
- 2) $C_{g,min} = \frac{C_{ox}C_{si,min}}{C_{ox}+C_{si,min}}$ where, $C_{si,min,dep} = \frac{\varepsilon_s}{W_{d,max}}$

Since, $W_{d,max} = \sqrt{\frac{4\varepsilon_s \phi_{Bp}}{qN_A}}$, Ideally we can find $N_A$!

But this estimate is NOT reliable!

$C_{min}$ does not coincide with $V_{th}$

Influence of $n_s$ - How to extract $N_A$?

3) Extracting $N_A$: Depletion ($N_A$-) – no influence of $n_s$

$$\frac{1}{C^2} = \left(\frac{1}{C_{ox}} + \frac{1}{C_{Si,dep}}\right)^2 = \left(\frac{1}{C_{ox}} + \frac{W}{\varepsilon_s}\right)^2$$

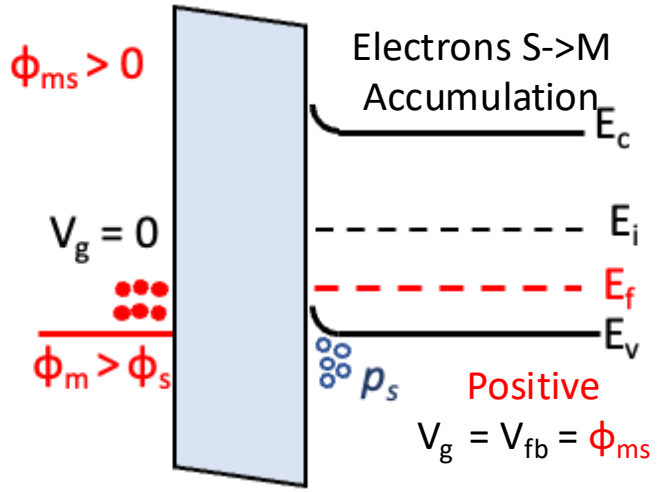
Differentiating with respect to $V_G$:

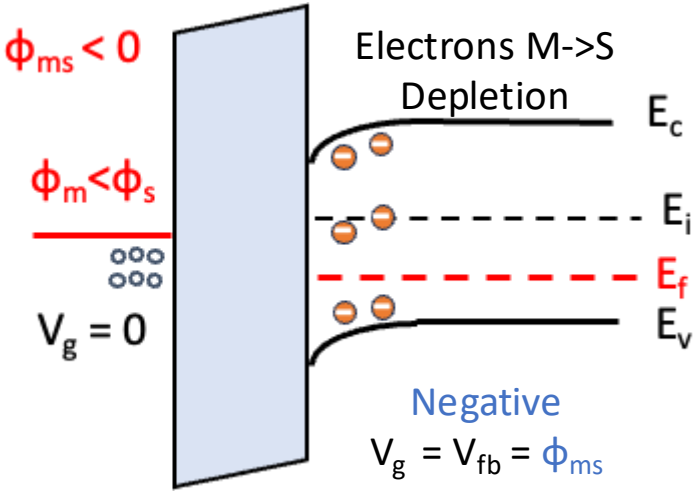
$\frac{d(1/C^2)}{dV_G} = \frac{2}{q\varepsilon_s N_A}$ {Easy Derivation: DIY!)

# MOSCAP CV non-idealities (1) -> $V_{fb} \neq 0$

Till now, $V_{fb} = 0$ @ $V_g = 0$ $Q_{si,m} = 0$ (a) $\phi_{ms} = 0$ (b) $Q_{oxide} = 0$

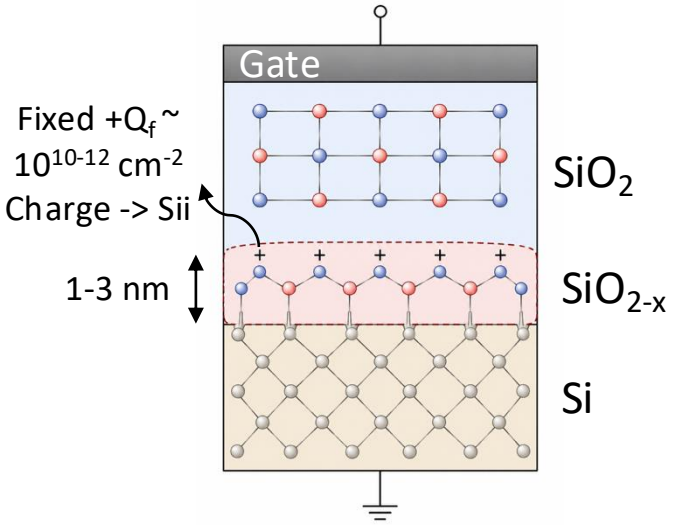
## (a) $\phi_{ms} \neq 0$

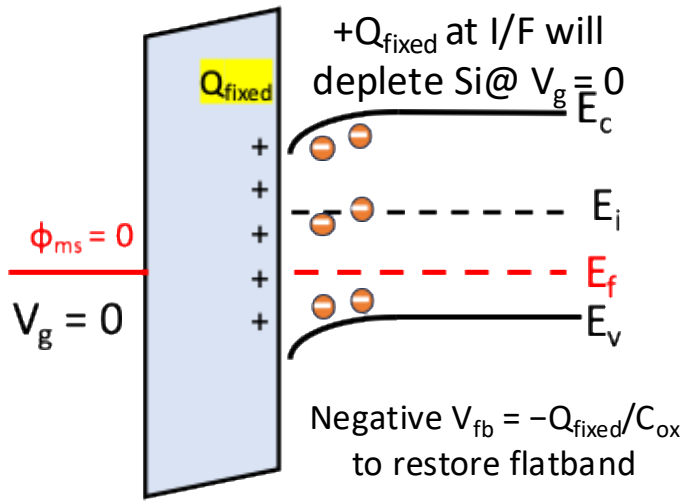



Restoration of flat band @$V_{gs} = \phi_{ms}$

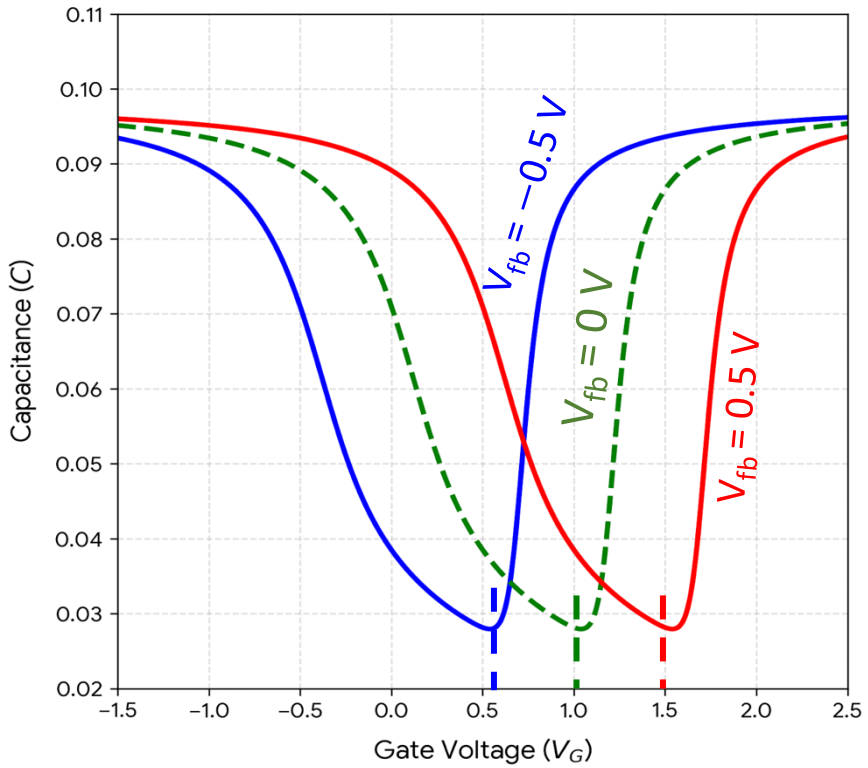
## (b) $Q_{oxide} \neq 0$




## Impact on CV: Lateral shift

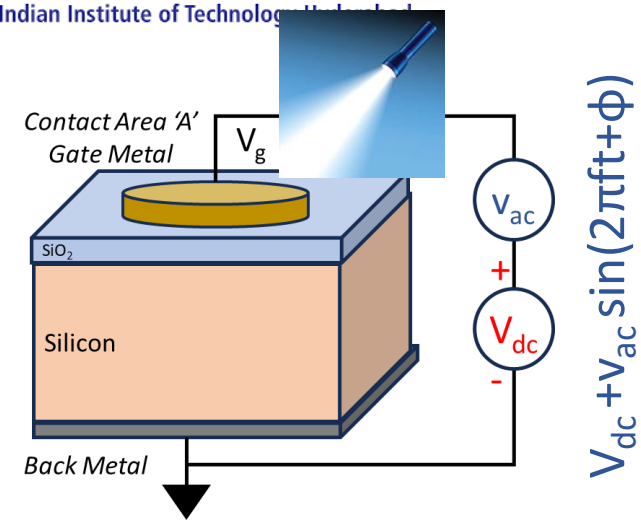
$$V_{fb} = \phi_{ms} - \frac{Q_f}{C_{ox}}$$



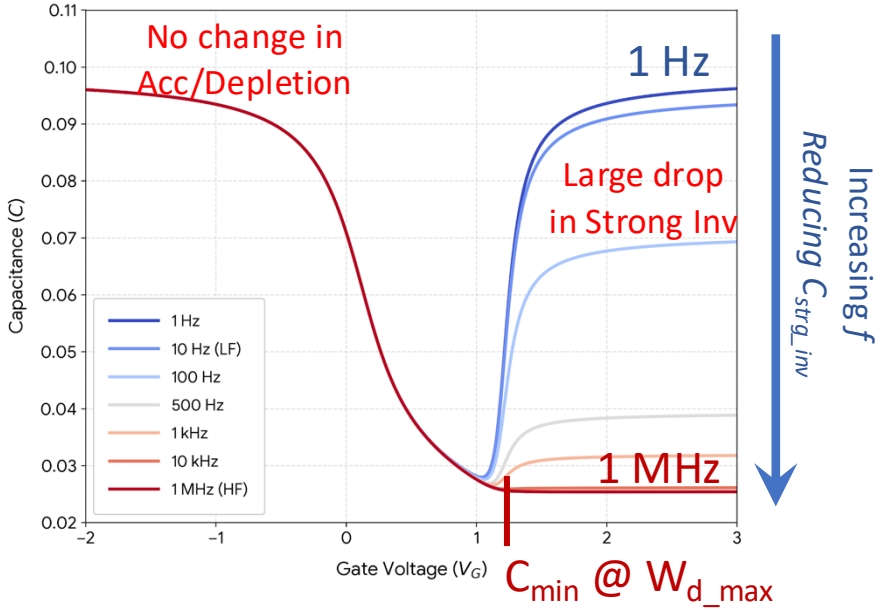

Notice $V_{th}$ also shifts by $V_{fb}$

$$V_{th} = V_{fb} + 2\phi_{Bp} + \frac{\sqrt{4q\varepsilon_s N_A \phi_{Bp}}}{C_{ox}}$$

# MOSCAP CV non-idealities (2) -> Impact of AC frequency




$V_{dc}+v_{ac}\sin(2\pi ft+\phi)$

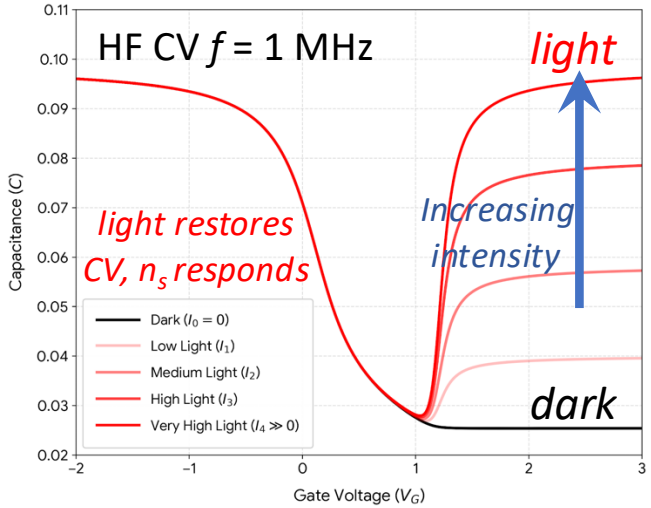


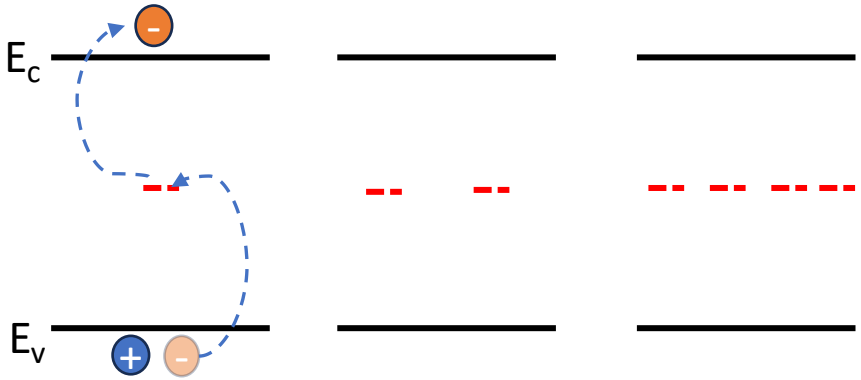

$C_{min}$ @ $W_{d_max}$

Can you spot 1 advantage HFCV $C_{min}$ at $V_{th}$ & $W_{dmax}$ -> Extract $N_A$

~~As if~~ respond –$n_s$ inversion layer "does not ~~exist~~" at high $f$ – *Why?*

MOSCAP CV with illumination




Pristine Si: $\tau_g$ ~ 1-10 ms

Med. Defects: $\tau_g$ ~ 1-10 µs

High defects $\tau_g$ ~ 1-10 ns

Light: generation of minority carriers: $n_s$!

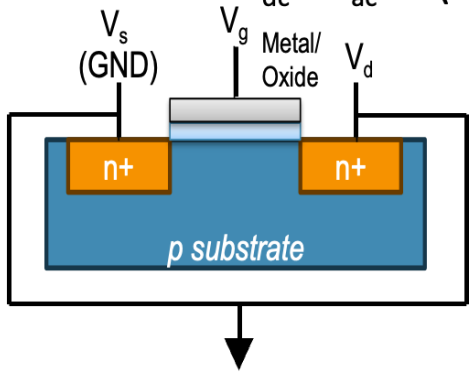
Minority generation lifetimes ($\tau_g$) vs. SRH Generation *via* mid-gap defects ($N_{trap}$)

$\tau_g \propto 1/N_{trap}$

Therefore, better quality of Si -> slower the freq response!

***MOSFET operate ONLY v. low f***

$V_{dc} + v_{ac}\sin(2\pi ft)$

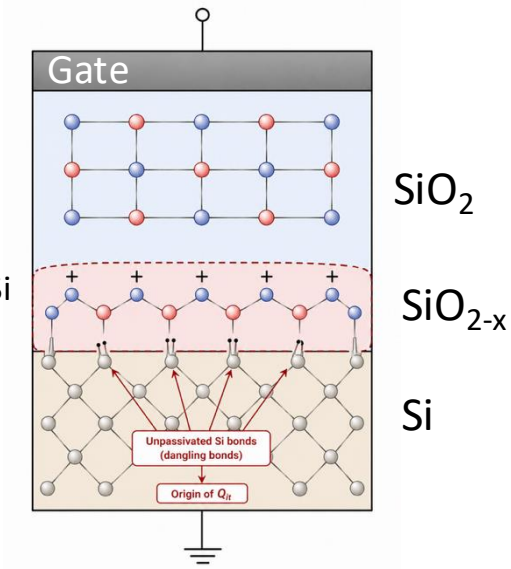


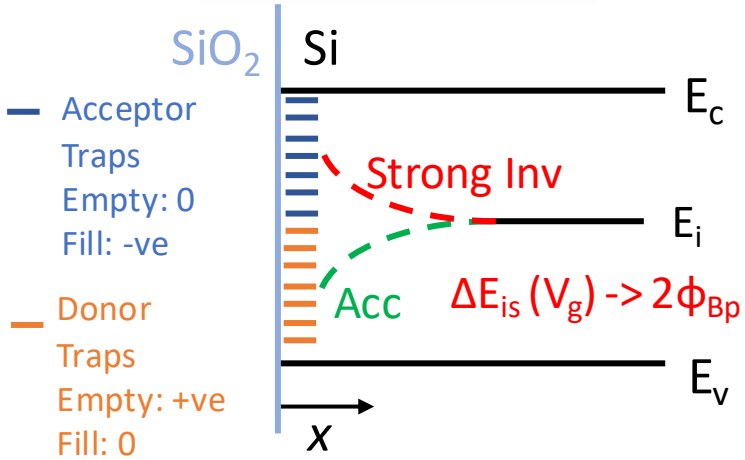

HW Q: MOSFET CV?

# MOSCAP CV non-idealities (3) -> Impact of Interface Traps

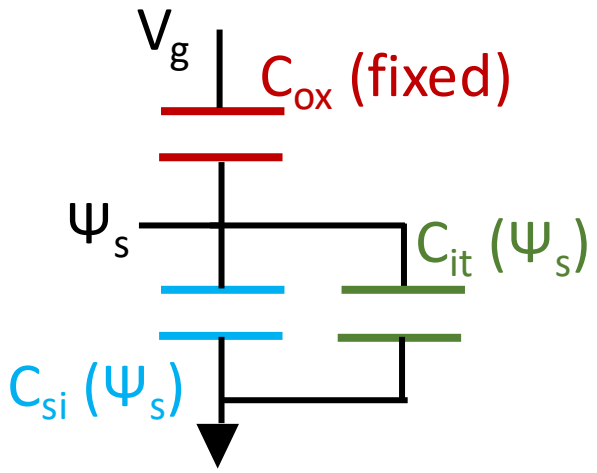
Physical Origin of $Q_{it}$
Dynamic : Fills empty $E_f$-$E_{is}$

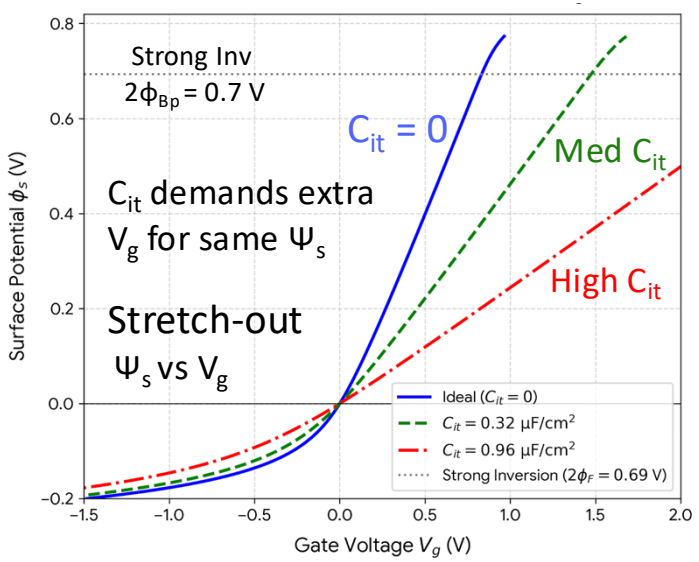
Abruptly Terminated Si Unsatisfied bonds *(right at I/F)*



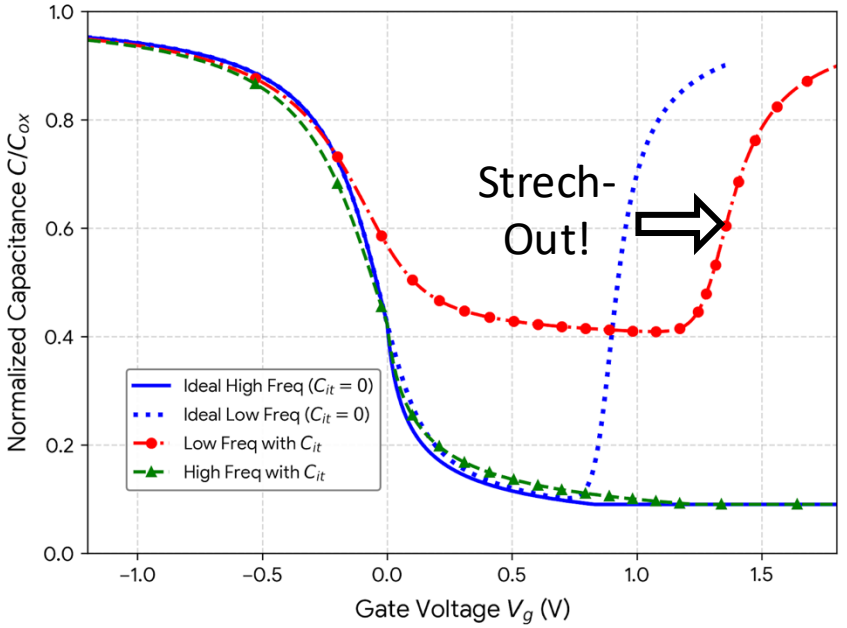




This "storage" of $Q_{it}$ ($E_{is}$, $V_g$) modelled -
$C_{it} = dQ_{it}/d\Psi_s$ Therefore, $C_{it}$ || $C_{si}$




Same Stretch-out will be seen in CV!




How to estimate $C_{it}$, $Q_{it}$ ?
*f* trick! High *f* $C_{it}$ won't respond!

$$C_{LF} = \frac{C_{ox}(C_{si} + C_{it})}{C_{ox} + C_{si} + C_{it}}$$

$$C_{HF} = \frac{C_{ox}C_{si}}{C_{ox}+C_{si}}$$

(almost like ideal)